Spectral World Renderer: Unified 3D Hyperspectral Unmixing

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https://ming053l.github.io/Spectral-World-Renderer/

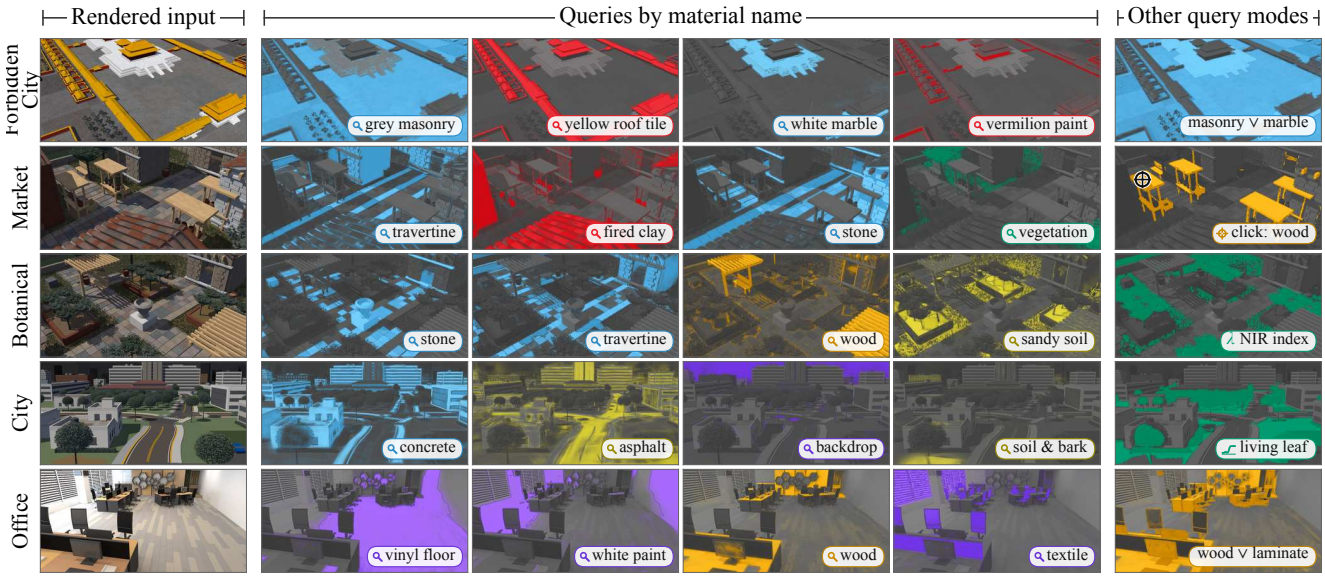

Figure 1. **From hyperspectral worlds to queryable material layers.** *Input:* multi-view hyperspectral images of worlds with known material composition, produced by the SWR world generator (left). *Output:* a Gaussian scene whose primitives carry material abundances and spectra. *Query:* a user selects a material by name, a click (crosshair), a reference spectrum, a band index or a combination, and SWR colours the retrieved surfaces in any view; each tile shows one query on a test view.

Abstract

Semantic models recognize what a 3D scene contains, but not what it is made of. In 3D, hyperspectral unmixing is applied to rendered images afterwards or tied to slow implicit fields, and multi-view data lack the composition needed to verify it. We present the Spectral World Renderer (SWR), which unifies world generation, rendering and unmixing. Its world generator lets a coding agent turn reference images and text into executable scene programs; measured spectra and shared rendering samples yield aligned hyperspectral observations, material labels and sub-pixel proportions, so decompositions are verified against known composition in the eight indoor and outdoor worlds of SWR-Bench. SWR unmixes inside rendering: it decodes mixtures over a shared endmember dictionary at every ray intersection of a compact Gaussian representation, so one pass renders spectra and queryable abundances, and a multi-view appearance regularization assigns endmembers to distinct appearances without material labels. It has the best mIoU in all eight worlds and the best mean PSNR on real captures, trains 3.7–12.9× *faster than unmixing baselines on City, renders 34.9 labelled views/s and separates RGB metamers.*

1. Introduction

Knowing what a scene is made of, not only what it contains, matters for robotic sorting, agriculture and building inspection. Foundation models such as CLIP, SAM and DINOv2 recognize and segment objects from RGB images, and their features can be lifted into 3D for open-vocabulary queries [36, 39, 58, 64]. They remain weak at material identification for three reasons: (1) three colour channels cannot separate surfaces of similar colour but different composition, such as a green-painted panel and a leaf; (2) their labels name objects rather than materials, and one object may be built from wood, laminate or metal; and (3) they assign one category per pixel, as do RGB material segmenters [4, 47, 80], whereas materials mix within a pixel.

Hyperspectral imaging supplies the missing evidence: the shape of a reflectance spectrum over tens to hundreds

of bands characterizes the material [19, 21, 40, 51], and spectral unmixing explains each spectrum as a non-negative mixture of material signatures, or endmembers [5]. Extended to 3D, unmixing would reveal which materials each surface point contains, consistently across views.

This extension is difficult. (1) Hyperspectral data have many bands, so representations that store a spectrum per primitive grow with the band count. (2) Mixing occurs twice: within a pixel, as in 2D, and along each ray; unmixing rendered spectra afterwards cannot separate what compositing has mixed. (3) The decomposition is ambiguous: illumination changes the brightness and shape of a spectrum, and a model can explain two similar materials with one endmember while intensity absorbs their difference.

Existing methods treat reconstruction and unmixing as separate steps. (1) Spectral radiance fields and Gaussian splatting [9, 23, 45, 53, 56, 73, 79] reconstruct spectra, often with per-band quantities or codes on every primitive, and are evaluated mainly by spectral fidelity; materials are then estimated by unmixing their renderings. (2) UnMix-NeRF [60] unmixes inside a radiance field, but its implicit field makes training and rendering slow and lacks queryable primitives. Moreover, as Table 1 shows, the data needed to verify a decomposition are missing: (1) real multi-view hyperspectral datasets are tabletop or turntable captures of a few objects [9, 38, 42]; (2) they record radiance, but no material labels or sub-pixel proportions, which cannot be annotated reliably by hand; and (3) synthetic RGB datasets contain geometry and semantics, not spectra.

We therefore generate worlds whose composition is known by construction. In the world generator of our Spectral World Renderer (SWR), shown in Figure 2, a coding agent turns reference images and text into executable scene programs that fix geometry, materials, illumination and cameras. Measured spectra ground the materials, and shared sub-pixel samples render aligned RGB and hyperspectral observations, material labels and proportions. SWR-Bench offers eight worlds that separate fidelity from identity.

Spectral World Renderer unifies unmixing with rendering in Gaussian splatting and targets each difficulty. (1) An RGB-trained anchor provides compact geometry, and each Gaussian stores a band-independent feature instead of a spectrum, so storage does not grow with the bands. (2) A shared decoder maps this feature, at every ray intersection, to abundances over a scene-wide endmember dictionary, so mixing is resolved before compositing and one render yields both. (3) A separate intensity absorbs shading, and a multi-view appearance regularization keeps it from merging materials by assigning over-complete endmembers to distinct appearances without material labels. With abundances on every Gaussian, a material is queried by name, click, reference spectrum or band in 3 ms. SWR has the best mIoU in every world and trains 3.7–12.9× faster than unmixing baselines on City. Our contributions are:

Table 1. **Multi-view datasets with ground truth.** ID: material labels; Prop.: geometric area proportions; Gen.: worlds on demand. NeSpoF-Seg.: synthetic segmentation truth from UnMix-NeRF [60]; SWR-Bench also provides jointly sampled proportions and spectral variability.

Dataset	Bands	nm	Scenes ×views	Capture	ID	Prop.	Gen.
Replica	3	RGB	18	indoor	✗	✗	✗
Hypersim	3	RGB	461×168	indoor	✗	✗	✗
Kubric	3	RGB	–	random	✗	✗	✓
NeSpoF	21	450–650	4×18–30	tabletop	✗	✗	✗
NeSpoF-Seg.	21	450–650	4 scenes	synthetic	✓	✗	✗
HS-NeRF	128/141	371–1087	8×48–433	turntable	✗	✗	✗
HSI-SC	204	397–1004	3×60	turntable	✗	✗	✗
SWR-Bench	64	420–980	8×96–240	in/outdoor	✓	✓	✓

- The SWR world generator and SWR-Bench: a coding agent writes executable worlds whose shared sub-pixel samples render spectra, material labels and proportions, yielding eight worlds with measured spectra that make 3D decompositions verifiable.
- Spectral World Renderer, a compact Gaussian representation that decodes mixtures at every intersection, so one render yields spectra and queryable 3D abundances with band-independent storage.
- A multi-view appearance regularization that clusters training pixels across views and binds each cluster to its own endmember, without material labels.
- Real-time material queries by name, click, reference spectrum or band, consistent across views, that separate RGB metamers and resist shadow.

2. Related Work

Hyperspectral 3D reconstruction. Radiance fields use neural or grid representations [2, 54, 55]; 3DGS uses alpha-composited Gaussians [35], with later quality and efficiency improvements [24, 30, 50, 68, 87]. Foundation-model features [39, 64] support object queries [8, 36, 63, 85, 91] and RGB material selection [16], and neural harmonic textures provide within-primitive appearance [12]. Hyperspectral fields render wavelength-dependent radiance [9, 38, 42, 45]; Gaussian counterparts store spectral quantities per primitive [23, 53, 56, 73, 79]. RGB inverse rendering estimates reflectance and illumination [6, 48, 89] or reflection components [81], rather than spectral endmembers.

Spectral unmixing. Unmixing estimates endmembers and non-negative abundances [5, 25, 37] through geometric extraction [57, 83], factorization or autoencoders [44, 59, 76], with spatial regularization [33] and spectral-variability or nonlinear models [26, 27]. Abundance maps support remote sensing and material analysis [19, 62]. UnMix-NeRF extends unmixing from 2D images to a radiance field [60]. Like it, we learn endmembers and abundances jointly, but on explicit Gaussians and before compositing.

Programmatic world generation. Procedural rules [65, 66], language-model layouts [15, 84] and scene code [29,

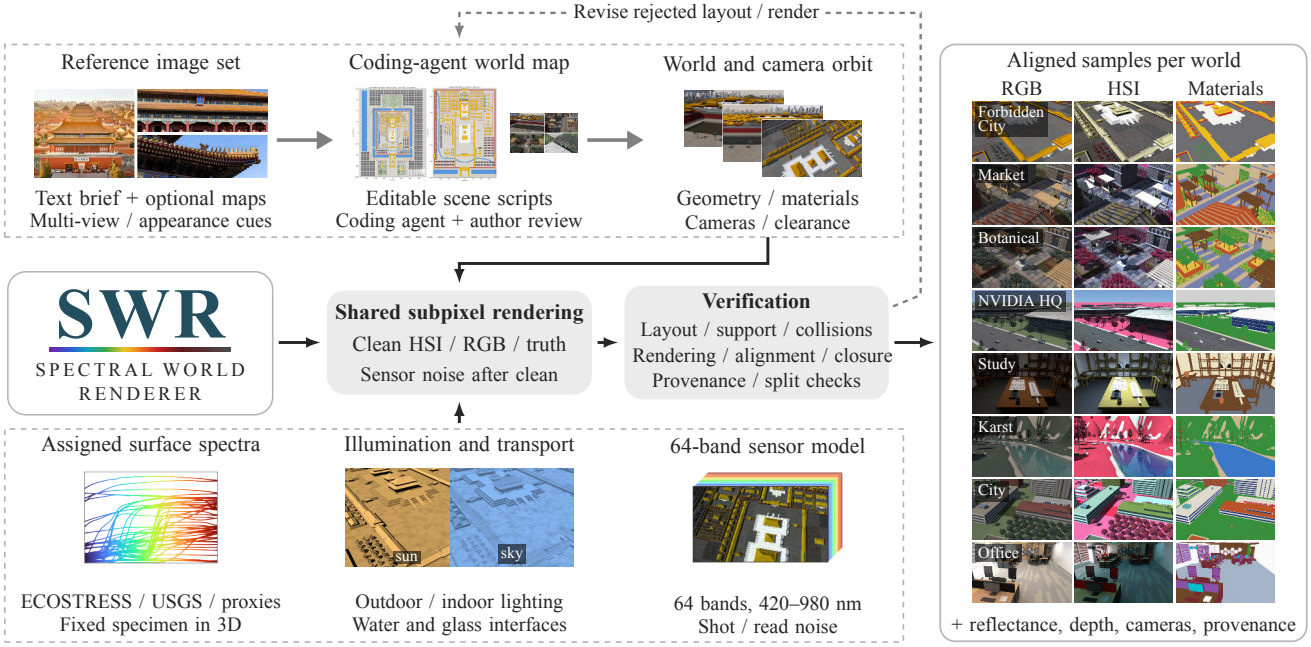

Figure 2. **The SWR world generator.** From reference photographs and a short brief, a coding agent writes an executable world whose every surface carries a library-measured spectrum. One shared sub-pixel rendering then produces the hyperspectral and RGB observations together with the material of every pixel, so the ground truth is exact for the rendered world, and checks on layout and rendering return flawed worlds for revision before release. This yields aligned, verifiable samples across all eight worlds (right) for scoring methods.

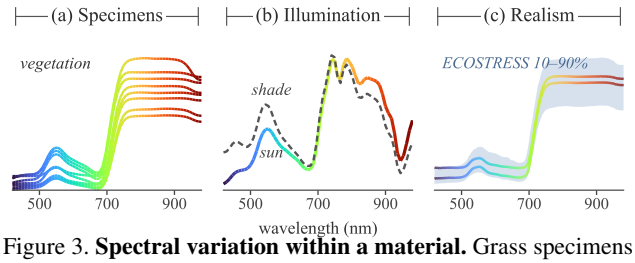

Figure 3. **Spectral variation within a material.** Grass specimens vary (a), and the same grass changes shape between sunlight and shade (b). Both broadleaf spectra shown lie within the range of independent leaf measurements (c).

46] generate 3D worlds. Programmable and generative world renderers couple scene state or G-buffers to video synthesis [31, 32, 49]. The SWR world generator binds measured spectra and declared transport to inspectable geometry, jointly sampling observations and composition.

Synthetic benchmarks and spectral simulation. Synthetic and scanned datasets supply geometry, semantics or intrinsic images [22, 70, 75]; game and agent benchmarks add G-buffers and checkable scene code [18, 32, 71, 86]. Spectral renderers and remote-sensing simulators trace wavelength-dependent transport [20, 34, 61, 82], and ray-traced vegetation scenes supply known abundances for 2D unmixing [13]. SWR-Bench adds shared sub-pixel sampling of images and area proportions, specimen and illumination variation, and stored coefficients for recomputation, as Table 1 summarizes, so every target can be regenerated.

3. Spectral World Renderer

Spectral World Renderer unifies 3D reconstruction and unmixing: one Gaussian representation, composited with one set of visibility weights, renders spectra, abundances and material queries at any band count, and one set of camera samples renders the observations and their ground truth. The world generator and SWR-Bench (Sec. 3.1) supply indoor and outdoor worlds, labels and sub-pixel proportions and measured spectra; the renderer (Sec. 3.2) unmixes with band-independent features, decoding before compositing and a separate intensity with multi-view appearance regularization (Sec. 3.3). Training uses no material labels; validation labels only select the number of endmembers K.

A scene is observed from V training cameras. Camera v provides RGB $\mathbf{I}_v \in [0,1]^{H\times W\times 3}$ and a co-registered hyperspectral cube $\mathbf{H}_v \in \mathbb{R}^{H\times W\times L}$ with L bands. For pixel p, $\mathbf{r} = \mathbf{r}_v(p)$ indexes its observed values $\mathbf{I}(\mathbf{r}) \in \mathbb{R}^3$ and $\mathbf{H}(\mathbf{r}) \in \mathbb{R}^L$. At a novel camera, the model renders a spectrum $\hat{\mathbf{H}}(\mathbf{r})$ and abundances $\hat{\mathbf{A}}(\mathbf{r})$ over shared endmembers.

3.1. World Generator and SWR-Bench

Scoring an unmixing requires per-pixel material proportions that real captures cannot provide. The SWR world generator therefore renders worlds whose images, material labels and area proportions share camera samples, so every target follows from the generated scene (Figure 2).

Reference-guided world generation. Users provide reference images, available maps and a text brief. A coding agent turns these cues into executable scene programs: parameter-

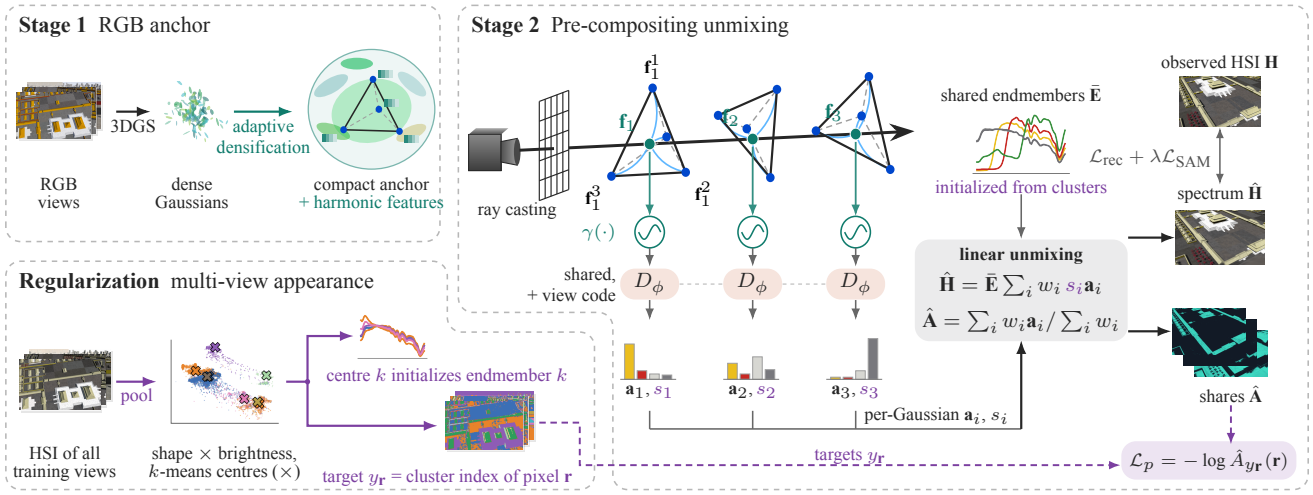

Figure 4. **Overview of Spectral World Renderer.** An RGB anchor fixes the geometry, and each Gaussian carries harmonic features whose size does not depend on the number of bands. At every ray intersection, a shared decoder maps the interpolated feature to abundances and an intensity, which mix scene-wide endmembers before compositing; the same visibility weights render the spectrum and the abundances. Clusters of multi-view appearance initialize the endmembers and regularize their assignment, so no material label enters training.

ized geometry and placement rules encode the spatial layout, while materials, lighting and camera trajectories remain editable. Rendered previews are reviewed by the agent and the authors for support, collisions and camera clearance. SWR-Bench comprises six outdoor sites and two interiors.

Shared-sample ground truth. Separate image and label renders disagree at silhouettes and mixed pixels. We therefore share S sub-pixel camera samples between them: for the pixel indexed by ray $\mathbf{r}$, $\mathbf{x}_n$ is the first surface hit of sample n and $m(\mathbf{x}_n)$ its material class. A Lambertian specimen j with measured reflectance $\rho_j(\lambda)$ at wavelength λ has, under direct or sky illumination u, band radiance

$$v^u_{j,\ell} = \int \psi_\ell(\lambda)\, \rho_j(\lambda) \Phi_u(\lambda)/\pi \, \mathrm{d}\lambda, \quad u \in \{\mathrm{d}, \mathrm{s}\}, \qquad (1)$$

where ψ_ℓ is the response of band ℓ and $\Phi_\mathrm{d}, \Phi_\mathrm{s}$ are direct and sky irradiance. The vectors $\mathbf{v}^u_j \in \mathbb{R}^L$ and accumulated shading coefficients $C^u_j(\mathbf{r})$ give

$$\begin{aligned} \mathbf{H}_{\text{clean}}(\mathbf{r}) &= \textstyle\sum_j (C^\mathrm{d}_j \mathbf{v}^\mathrm{d}_j + C^\mathrm{s}_j \mathbf{v}^\mathrm{s}_j), \\ A_c(\mathbf{r}) &= \tfrac{1}{S} \textstyle\sum_{n=1}^{S} [m(\mathbf{x}_n) = c]. \end{aligned} \qquad (2)$$

Radiance and area proportions thus share one footprint; released proportions renormalize A_c over non-sky samples. Here c indexes ground-truth classes and k, below, learned endmembers. Beyond this Lambertian model, NVIDIA HQ adds rough specular highlights, a glass reflection and an analytic sky with haze, Karst water adds reflection, refraction and absorption, and the interiors are path traced in Mitsuba [34]; each effect is stored as a separate contribution.

Measured spectra and their variation. Reflectance comes from ECOSTRESS and USGS measurements [40, 51], and every assignment is recorded, so each observation can be traced to the spectra that produced it. Specimens differ within a class, and direct sunlight and skylight change the spectral shape of one surface, as shown in Figure 3(a,b), so no single spectrum describes a whole material; each surface keeps one specimen across views. Outdoor worlds use ASTM G173 sunlight and skylight [1], interiors white sources [11]. We render 64 bands from 420 to 980 nm, clean and with shot and read noise [17].

Verification. Each world is checked for label alignment and integrity, spectral provenance and train/test separation. Stored coefficients and separate contributions re-compose the clean outdoor radiance, indoor transport is checked before noise is added, and the assigned spectra fall within independent library measurements, as Figure 3(c) shows.

3.2. Unmixing at Every Ray Intersection

Unmixing a rendered image afterwards acts on spectra that compositing has already mixed. Spectral World Renderer instead unmixes inside rendering: at every ray intersection, a shared decoder turns the local feature into abundances and an intensity, and compositing accumulates both, so one render yields spectra and abundances, as Figure 4 shows.

RGB anchor and harmonic features. Hyperspectral observations carry sensor noise and illumination variation, which across many views also degrade the recovered geometry. We therefore fit a 3DGS anchor to co-registered RGB observations, supplied with the data or derived from visible-wavelength responses; through this natural interface, advances in RGB Gaussian splatting transfer directly. FastGS densification and pruning [68] retain only primitives supported across views, which accelerates rendering and leaves fewer features to optimize and fewer intersections to unmix. Unlike HyperGS [79], the anchor requires no spectral autoencoder. It has N Gaussians G_i with mean $\boldsymbol{\mu}_i$, rotation $\mathbf{R}_i$, scale $\boldsymbol{\sigma}_i$ and opacity o_i; after fitting, we drop colours and fold the rasterizer's screen-space variance into the scales. We attach a neural harmonic texture [12] to

each Gaussian, whose in-primitive variation lets few Gaussians represent high-frequency detail: four feature vectors $\mathbf{f}_{im} \in \mathbb{R}^d$, $m = 1, \dots, 4$, with $d = 12$. A ray $\mathbf{r}$ queries G_i at its point of maximal response, whose barycentric coordinates $\beta_{im}(\mathbf{r})$ in a tetrahedron interpolate the vertex features:

$$\mathbf{f}_i(\mathbf{r}) = \sum_{m=1}^{4} \beta_{im}(\mathbf{r}) \mathbf{f}_{im} \in \mathbb{R}^d. \tag{3}$$

Thus different rays through one Gaussian access different features. We refine these features and geometry with the NHT RGB shader, without further densification, then freeze geometry for hyperspectral learning. Each Gaussian stores 11 geometry and 48 feature values, independent of L.

Pre-compositing material decoder. For each intersection, the feature in Eq. 3 supplies the spatial input to a shared MLP D_ϕ. We encode it as $\gamma(\mathbf{f}) = [\sin \mathbf{f}; \cos \mathbf{f}]$ [77] and supply a directional code $\mathbf{Y}(\mathbf{d}_i(\mathbf{r})) \in \mathbb{R}^9$: real spherical harmonics of degree ≤ 2 [67], where $\mathbf{d}_i$ points from the Gaussian centre to the camera. The decoder produces

$$\mathbf{z}_i(\mathbf{r}) = D_\phi(\gamma(\mathbf{f}_i(\mathbf{r})), \mathbf{Y}(\mathbf{d}_i(\mathbf{r}))) \in \mathbb{R}^{K+1}. \tag{4}$$

Here K is the number of shared endmembers. The first K outputs yield abundances $\mathbf{a}_i(\mathbf{r}) = \mathrm{softmax}(\mathbf{z}_i^{1:K}(\mathbf{r}))$; the last gives intensity $s_i(\mathbf{r}) = \mathrm{softplus}(z_i^{K+1}(\mathbf{r}))$. The MLP has three width-128 SiLU hidden layers; a zero-initialized linear map adds the directional code to layer one.

These coefficients mix a scene-wide dictionary $\bar{\mathbf{E}} = [\bar{\mathbf{e}}_1, \dots, \bar{\mathbf{e}}_K] \in \mathbb{R}^{L \times K}$. Each endmember is parameterized by $\boldsymbol{\eta}_k \in \mathbb{R}^L$ as

$$\bar{\mathbf{e}}_k = \frac{\mathrm{sigmoid}(\boldsymbol{\eta}_k)}{\| \mathrm{sigmoid}(\boldsymbol{\eta}_k) \|_2} \in \mathbb{R}^L_{\geq 0}. \tag{5}$$

Unit normalization assigns spectral direction to the dictionary and magnitude to s_i. These terms yield the intersection spectrum as a scaled linear mixture [5]:

$$\mathbf{h}_i(\mathbf{r}) = s_i(\mathbf{r}) \bar{\mathbf{E}} \mathbf{a}_i(\mathbf{r}) \in \mathbb{R}^L_{\geq 0}. \tag{6}$$

Sec. 3.3 initializes and regularizes the dictionary.

Joint spectral and mixture rendering. Order intersected Gaussians from front to back, and let $\alpha_i(\mathbf{r})$ be their effective opacity. Transmittance $T_i(\mathbf{r}) = \prod_{i'<i}(1 - \alpha_{i'}(\mathbf{r}))$ gives visibility weights $w_i(\mathbf{r}) = T_i(\mathbf{r})\alpha_i(\mathbf{r})$ [35]. Using the same weights for spectra and coefficients yields

$$\hat{\mathbf{H}}(\mathbf{r}) = \sum_i w_i(\mathbf{r}) \mathbf{h}_i(\mathbf{r}), \quad \hat{\mathbf{A}}(\mathbf{r}) = \frac{\sum_i w_i(\mathbf{r}) \mathbf{a}_i(\mathbf{r})}{\sum_i w_i(\mathbf{r})}. \tag{7}$$

Each $\mathbf{a}_i$ satisfies the abundance non-negativity constraint (ANC) and the abundance sum-to-one constraint (ASC) of linear unmixing [25], and for a non-empty ray $\hat{\mathbf{A}}$ is a convex combination of them and satisfies both. Intensity scales the rendered spectrum but not the rendered abundances: $\hat{\mathbf{H}} = \bar{\mathbf{E}} \sum_i w_i s_i \mathbf{a}_i$, whereas $\hat{\mathbf{A}}$ uses w_i alone. Decoding before this accumulation retains a mixture per intersection, which supports 3D queries on each Gaussian's abundance averaged over its training rays (Sec. 4.4).

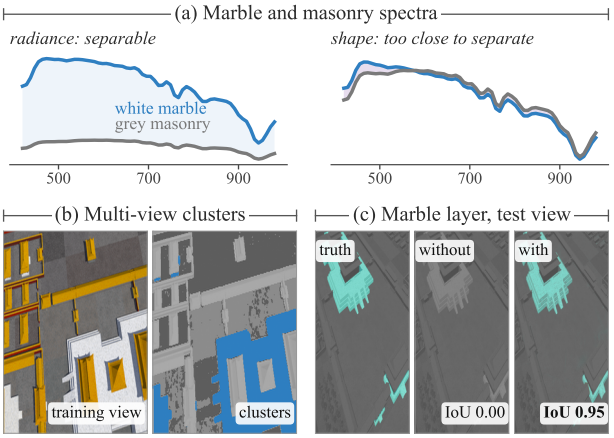

Figure 5. **Multi-view appearance regularization** (Forbidden City). (a) White marble is four times brighter than grey masonry but only 3.8° apart in spectral shape; since intensity absorbs brightness, one endmember can explain both. (b) Clustering the pixels of all training views by shape and brightness separates them, and every view shares the same clusters. (c) Kept as a weak regularizer during training, these clusters give marble its own layer; without them it is absorbed into masonry.

3.3. Multi-View Appearance Regularization

Assignment ambiguity. Eq. 6 fits each spectrum but does not determine which endmember explains it. Spectrally similar materials may collapse onto a single endmember while the intensity s absorbs their brightness difference, analogous to the scaling factors of extended mixing models [14, 27]: the spectra are reconstructed accurately, yet the materials become indistinguishable in $\hat{\mathbf{A}}$. We therefore derive an initial endmember for each appearance from the multi-view training data and retain this assignment as a weak regularizer, illustrated in Figure 5.

Over-complete dictionary. Since one material spans several spectra (Figure 3), we follow endmember bundles and multiple-endmember unmixing [3, 69, 74] and choose K larger than the number of materials, one endmember per appearance mode. More endmembers alone do not decide which endmember serves which surface; without guidance they drift to dominant materials, as in Figure 5(c).

Multi-view appearance clusters. We pool the foreground pixels of all training views and describe each 3×3-smoothed spectrum $\tilde{\mathbf{H}}(\mathbf{r})$ by its shape and brightness:

$$\mathbf{q}_\mathbf{r} = \left[\frac{\tilde{\mathbf{H}}(\mathbf{r})}{\| \tilde{\mathbf{H}}(\mathbf{r}) \|_2}, \; \kappa \, b(\tilde{\mathbf{H}}(\mathbf{r})) \right], \tag{8}$$

where b is the log norm of the spectrum, standardized robustly over the training views and clipped. The bounded weight κ keeps brightness secondary to shape, yet brightness still separates surfaces whose spectra are nearly parallel and which the spectral angle alone would merge. Farthest-point seeding followed by k-means groups these descriptors into K clusters, whose assignments serve

Table 2. **Material identification without training labels and spectral fidelity on SWR-Bench.** Spectral World Renderer identifies materials best or on par in every world and learns the endmembers closest to the true spectra; a lower spectral angle (3DGS in Market and Study) does not imply better identification. Rare: classes under 3 % of test pixels; Spectra: specimens; EM in degrees.

					UnMix-NeRF [60]				3DGS [35] + VCA / NNLS				Spectral World Renderer			
Scene	Setting	Classes	Rare	Spectra	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓
Forbidden City	outdoor	9	3	47	26.13	7.37	0.19	27.4	25.03	5.05	0.17	30.7	28.37	4.45	0.55	4.4
Market	outdoor	6	0	29	31.87	3.61	0.13	30.9	30.02	1.85	0.24	26.9	33.63	1.95	0.78	2.4
Botanical	outdoor	6	0	29	30.24	5.99	0.10	38.8	29.57	3.12	0.22	27.1	32.88	2.97	0.71	2.4
NVIDIA HQ	outdoor	8	2	27	18.29	14.98	0.08	43.0	22.27	12.31	0.13	24.4	25.30	8.25	0.31	3.7
Study	indoor	5	2	5	33.78	2.81	0.20	22.9	32.53	1.52	0.30	14.9	33.14	2.01	0.52	0.9
Karst	outdoor	5	2	10	26.91	3.29	0.38	25.8	27.19	2.26	0.35	28.8	28.10	2.15	0.73	2.9
City	outdoor	11	5	56	27.00	4.40	0.10	30.4	25.61	4.21	0.21	29.1	28.18	3.53	0.52	3.1
Office	indoor	8	3	35	25.33	6.65	0.11	18.8	25.01	5.03	0.17	41.3	27.54	4.31	0.53	3.3

as pseudo-labels in the spirit of clustering-based self-supervision [7]. Cluster centre k initializes endmember k, and each pixel receives its cluster as pseudo-label $y_{\mathbf{r}}$.

Persistent assignment regularization. During training, each pseudo-labelled pixel should concentrate its abundance (Eq. 7) on the endmember of its cluster:

$$\mathcal{L}_p = -\frac{1}{|\mathcal{B}_p|} \sum_{\mathbf{r}\in\mathcal{B}_p} \log \hat{A}_{y_{\mathbf{r}}}(\mathbf{r}), \tag{9}$$

where $\hat{A}_{y_{\mathbf{r}}}(\mathbf{r})$ is entry $y_{\mathbf{r}}$ of $\hat{\mathbf{A}}(\mathbf{r})$ and $\mathcal{B}_p$ holds the pseudo-labelled pixels of the batch whose rays hit the scene. Because a Gaussian is rendered into many training views, its abundances must agree with the pseudo-labels of all of them: like semantic lifting in radiance fields [72, 85, 90], the loss lifts 2D appearance clusters into a 3D assignment by multi-view agreement. It constrains which endmember explains a pixel, not the spectrum of that endmember or the intensity. Freezing endmembers for the first 3k updates, while the weight of $\mathcal{L}_p$ decays to its floor (Sec. 3.4), keeps initially distinct appearances from merging.

3.4. Optimization

Stage 1 fits the anchor and refines geometry and features on RGB with $\mathcal{L}_{\text{RGB}} = 0.8\mathcal{L}_1 + 0.2\mathcal{L}_{\text{D-SSIM}}$. Stage 2 freezes geometry and updates features, decoder and endmembers from a training camera and uniform pixel batch $\mathcal{B}$, with

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{rec}} + \lambda_{\text{SAM}}\mathcal{L}_{\text{SAM}} + \lambda_p(t)\mathcal{L}_p, \\ \mathcal{L}_{\text{rec}} &= \tfrac{1}{|\mathcal{B}|L} \textstyle\sum_{\mathcal{B}} \|\hat{\mathbf{H}}_{\mathbf{r}} - \mathbf{H}_{\mathbf{r}}\|_1, \\ \mathcal{L}_{\text{SAM}} &= \tfrac{1}{|\mathcal{B}_+|} \textstyle\sum_{\mathcal{B}_+} \text{SAM}(\hat{\mathbf{H}}_{\mathbf{r}}, \mathbf{H}_{\mathbf{r}}), \end{aligned} \tag{10}$$

where $\mathcal{B}_+$ has non-zero targets, SAM is in radians, $\lambda_{\text{SAM}} = .05$ and $\lambda_p(t) = \max(.005, .05(1 - t/3000))$ at update t.

4. Experiments

Data and baselines. SWR-Bench contains eight indoor and outdoor worlds. Material comparisons use UnMix-NeRF [60] and an L-channel 3DGS [35] with endmembers extracted by VCA [57] from training-view renders and NNLS [43]. For fidelity we also compare HyperGS [79] and DD-HGS [56]. Real data are four multispectral NeSpoF scenes [38], the 204-band HSI-SC apple [42] and five HS-NeRF scenes [9], spanning 21 to 204 bands.

Table 3. **RGB feature refinement.** Refining geometry and harmonic features on the RGB anchor gives the decoder spatially structured features and improves both identification and spectral fidelity; without it, the decoder reads random features.

	Market (SWR-Bench)			HSI-SC apple		City
	mIoU↑	PSNR↑	SAM↓	PSNR↑	SAM↓	train↓
w/o refinement	0.66	30.73	3.20	28.44	1.67	–
SWR	**0.78**	**33.63**	**1.95**	**30.99**	**1.34**	6 min 37 s

Protocol. Methods train on noisy views; PSNR and the spectral angle SAM [41] (mean and 95th percentile, in degrees) use clean truth on foreground pixels. Each endmember is mapped to the class holding most training pixels it dominates; with this map frozen, we report the dominant-endmember mIoU on test views. EM is the mean angle between each class's mean spectrum over pure training pixels and its learned endmember under one-to-one assignment.

Implementation details. The anchor trains for 30k iterations, feature refinement for 5k and the decoder for 30k Adam steps (Sec. C). Validation mIoU selects K per world from the class count, 16, 24 and 32 (K=24 for Market, NVIDIA HQ and Study, K=32 elsewhere); real captures use K=16, and baselines K equal to the class count. At K=24, 3DGS+VCA/NNLS stays below SWR in every world but Study, where the two tie (Table S6).

4.1. Material Identification and Abundances

Fidelity–identification gap. A lower SAM does not imply better labels, as Table 2 and Figure 6 show: on Market and Study, 3DGS reaches a lower SAM than SWR yet a far lower mIoU, since post-hoc unmixing operates on contributions already mixed by compositing. NNLS with the true spectra reaches 0.84–0.94 mIoU on clean radiance but 0.32–0.77 under noise (Table S2), so noise and the learned decomposition both limit identification.

4.2. Ablation and Analysis

RGB feature stage. RGB feature refinement on the fixed anchor takes 36.5 s of the 397 s City pipeline; refining ge-

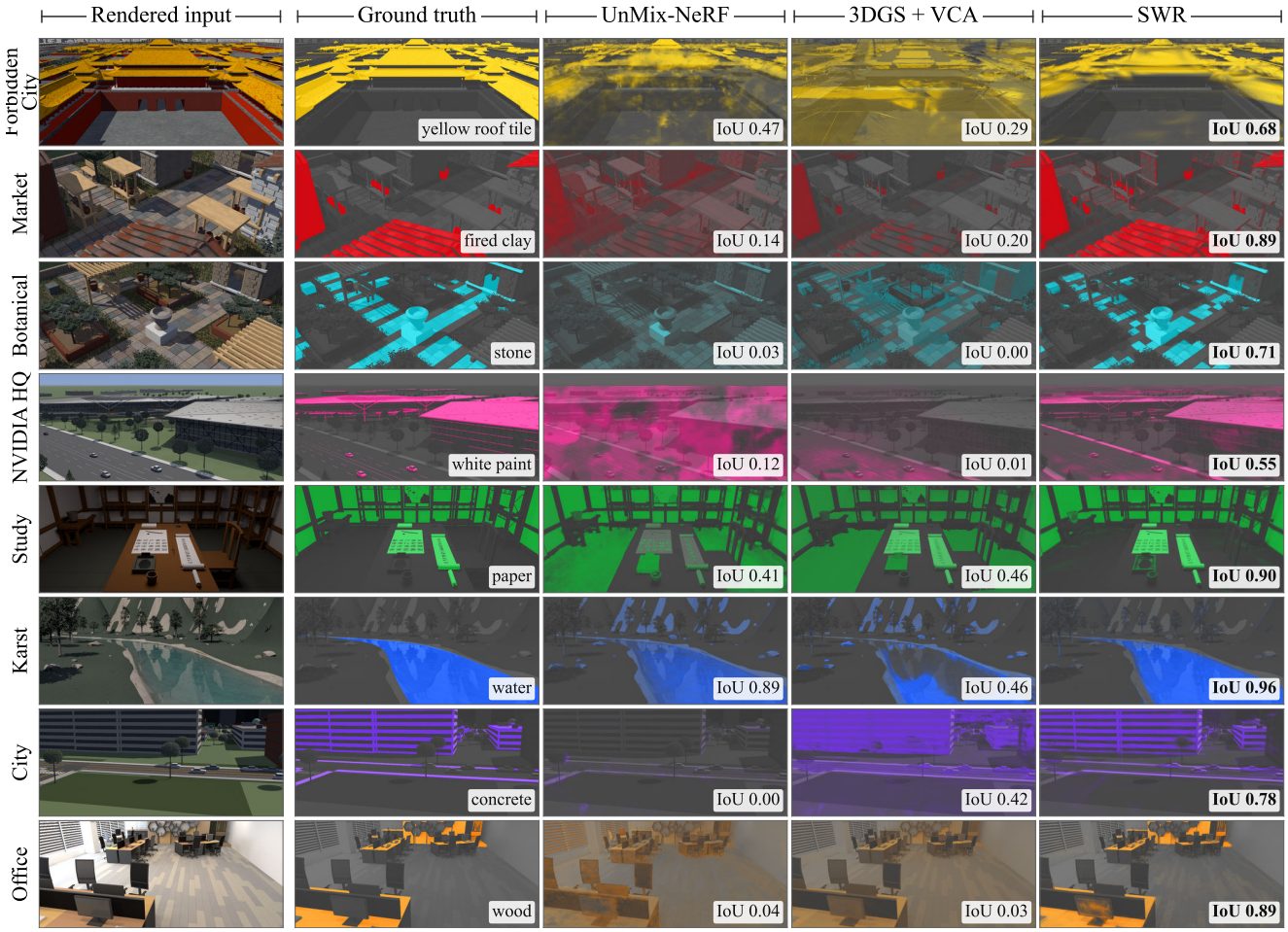

Figure 6. **Material layers across the eight worlds.** Each row tints one material by its abundance in a test view, for a material that every method maps an endmember to; IoU: summed abundance > 0.5 against the true mask. Unmixing rendered spectra afterwards (3DGS + VCA) or inside a radiance field (UnMix-NeRF) spreads the material over unrelated surfaces, whereas Spectral World Renderer recovers the layer closest to the ground truth. All rows use the models of Table 2.

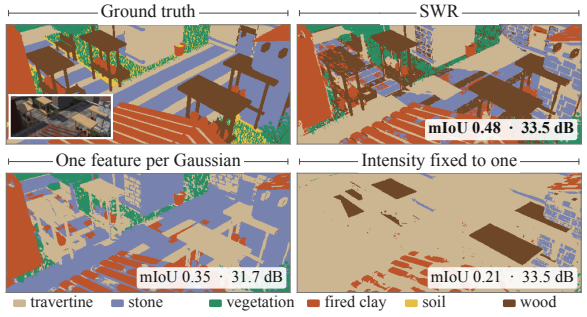

Figure 7. **Within-primitive variation** on a Market test view (inset: input; values for this view). One feature per Gaussian blends materials across edges and shadows.

ometry and feature layout, it improves Market by 2.9 dB and 0.12 mIoU and the real apple by 2.6 dB, as Table 3 shows.

Within-primitive variation. Figure 7 varies single components on Market without $\mathcal{L}_p$. A single feature per Gaussian costs 1.6 dB and lowers mIoU from 0.47 to 0.34, as a primitive across an edge has one mixture. Fixed intensity keeps PSNR but lowers mIoU to 0.22, as mixtures absorb shading.

Unmixing inside rendering. Decoding features after compositing instead raises Market PSNR from 33.72 to 34.37 dB yet lowers mIoU from 0.47 to 0.37 (Table S7), as features averaged along a ray blend surfaces, the gap post-hoc unmixing shows in Table 2.

Material allocation limits. Appearance regularization recovers Forbidden City marble (Figure 5(c)) but misses rare materials such as ink in Study, although NNLS with the true class spectra separates most of them (Figure S7).

4.3. Real Captures and Efficiency

Fidelity without material truth. SWR has the highest PSNR on each real dataset in Table 5. The largest margin is on the 204-band apple in Figure 8, where PSNR, mean SAM and p95 all improve over HyperGS. On HS-NeRF, 3DGS reaches a lower mean and p95 angle.

Efficiency. As Table 4 lists, the City pipeline lasts 6 min 37 s versus 24 min for UnMix-NeRF and 85 min for

Table 4. **Efficiency.** FPS: views/s with spectra and label maps. Ours: our optimized fused fp16 decoder, skipping hits of weight $< 10^{-4}$ (99.97 % identical labels); baselines: reference code. City training covers all four stages, from anchor to decoder.

	City	FPS↑ at $960 \times 540 \times 64$				
Method	train↓	Market	Bot.	Study	Karst	Mean
UnMix-NeRF	24 min 17 s	0.18	0.21	0.21	0.12	0.18
3DGS+VCA/NNLS	85 min 2 s	2.53	2.51	4.70	4.73	3.6
SWR	**6 min 37 s**	**36.5**	**39.3**	**36.3**	**27.4**	**34.9**

Table 5. **Real-source fidelity.** L: bands; p95: 95th-percentile SAM; means over 4 NeSpoF / 5 HS-NeRF scenes; baseline: best PSNR of 3DGS, HyperGS, DD-HGS.

		PSNR-leading baseline				SWR		
Source	L	Method	PSNR↑	SAM↓	p95↓	PSNR↑	SAM↓	p95↓
NeSpoF	21	HyperGS	28.36	4.58	9.44	**29.39**	**3.86**	**8.04**
Apple	204	HyperGS	27.95	1.64	4.88	**30.67**	**1.35**	**3.87**
HS-NeRF	128 141	3DGS	26.32	**5.71**	**16.75**	**26.51**	5.97	17.31

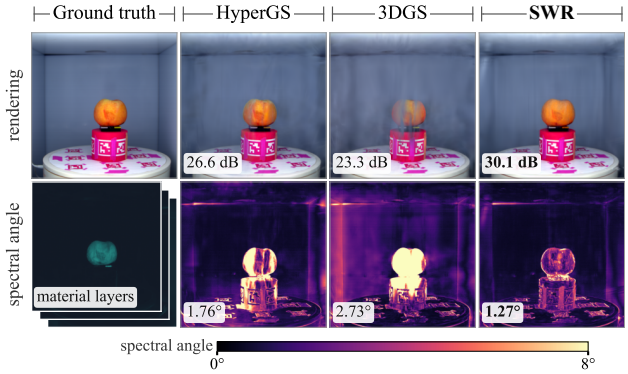

Figure 8. **204-band real capture (HSI-SC apple).** SWR renders the most faithful spectra in this view; inset: its abundances.

3DGS+VCA/NNLS, and our optimized fused fp16 decoder renders 34.9 views/s with label maps.

4.4. Application: Material Queries

Seeing beyond RGB. Chlorophyll absorbs red light and leaf tissue reflects near-infrared, a red edge near 700 nm that minerals and dry leaves lack. Foliage-coloured mineral leaves (2° apart in RGB, 42° in spectrum) and dry crowns thus receive their own endmembers, as Figure 9(b,c) shows. **Querying primitives directly.** In hyperspectral images, such materials are usually isolated by supervised classifiers [10, 28, 52, 78] or by unsupervised subspace clustering [88], which need labels or a run per image. SWR needs neither: with abundances on explicit, rasterizable Gaussians, a material query selects primitives, consistently across views. The mineral-leaf query retrieves no ordinary tree, whereas the RGB query retrieves eleven, as shown in Figure 9(a). A measured spectrum can also serve as the query: with sun and sky illumination factored out, a pixel clicked in shadow retrieves 77 % of the sunlit part of its material, against 10 % for RGB (mean over four outdoor worlds, 3 ms per query; Sec. I), as Figure 9(d) shows.

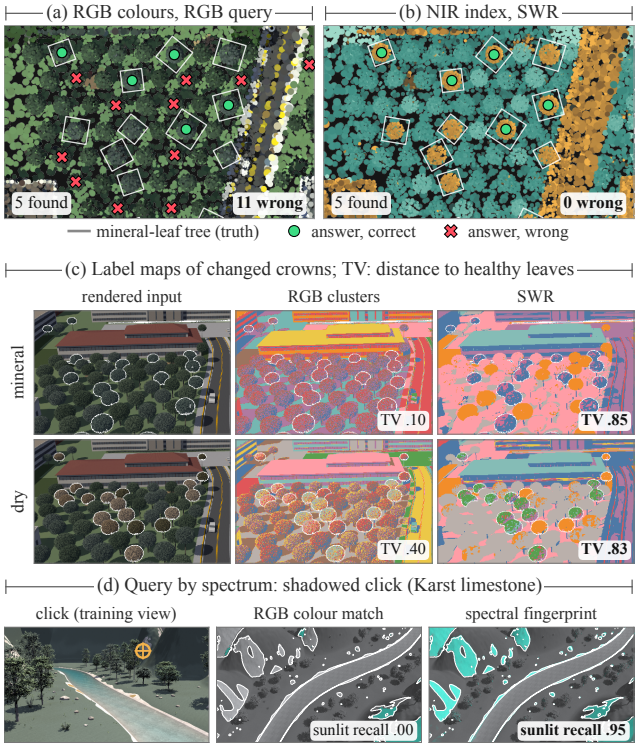

Figure 9. **Material queries on per-Gaussian abundances.** In City, foliage-coloured mineral leaves replace the boxed crowns. (a) On RGB colours, the same query also retrieves ordinary trees. (b) The mineral lacks the NIR rise of foliage, so the query on per-Gaussian abundances retrieves only these trees. (c) For mineral and dry leaves, SWR separates the changed crowns from healthy foliage, whereas RGB clusters mix them. (d) A spectrum clicked on shadowed limestone (Karst) also retrieves the sunlit rock, which RGB misses (recall of this click over all test views).

5. Conclusion

Spectral World Renderer brings world generation, rendering and unmixing into one verifiable framework for 3D hyperspectral analysis. Its world generator renders observations, labels and proportions from shared samples, so decompositions are scored against known composition. By decoding band-independent features into abundances at every ray intersection, SWR obtains spectra and abundances from a single render: it has the best mIoU in all eight worlds and the best mean PSNR on real captures, trains City in under seven minutes, renders 34.9 labelled views/s, and its per-Gaussian abundances and spectra answer queries by name, click, reference spectrum or band. Future work will enrich light transport and sensor models, allocate endmembers to rare materials and handle dynamic scenes.

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Supplementary Material for Spectral World Renderer: Unified 3D Hyperspectral Unmixing

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https://ming053l.github.io/Spectral-World-Renderer/

Contents. Figures, tables and equations carry the prefix S; unprefixed numbers refer to the main paper.

A SWR-Bench worlds and protocol 1
B Spectra of SWR-Bench 1
C Implementation details 2
D Decoder analysis 3
E Real captures 5
F Efficiency 5
G Rare materials and altered leaves 6
H Object queries 9
I Query by spectrum 9
J Additional views of SWR-Bench 12

A. SWR-Bench Worlds and Protocol

Generation and rendering. A coding agent builds each world from reference photographs, available maps and a written brief, revising it from rendered previews; the authors review the previews and approve each world, and the worlds, scene programs and code will be released (Table S1). Spectral images, RGB and ground truth share 16 sub-pixel samples per pixel, so labels and area proportions describe the first-hit surfaces that produce most of the radiance; transmitted and interreflected light is stored as separate contributions. Each surface keeps one measured specimen in world space across all views. Outdoor worlds use Lambertian surfaces under ASTM G173 sunlight and skylight [1]; NVIDIA HQ adds glossy highlights, glass reflection and an analytic sky, Karst adds refracting water with pure-water absorption, and the Study and Office are path traced in Mitsuba [4] with interreflection. All worlds are rendered at 960×540 with the same 64 Gaussian band responses over 420–980 nm.

Noise and protocol. Every world is released clean and with Poisson shot and Gaussian read noise scaled from the training views. All methods share the splits, classes and stereo point cloud. Each world has full primary-ray coverage, at least 99.9 % agreement between instance and material maps, and outdoor radiance that re-composes exactly from the stored coefficients.

Classes and City variants. Similar spectra are merged into one class, for example the stones of Market and Botanical and the paving and brick of Forbidden City. Four City variants change one factor each: foliage-coloured mineral leaves (*metamer*), dry leaves on 87 trees (*stress*), and a morning or afternoon sun. Every object keeps its instance and spectral sample, so Sec. H can score the object queries.

Metrics. PSNR and spectral angle are computed against the clean radiance on the ground-truth support. Endmembers are mapped many-to-one to classes by majority vote on training views; with the map frozen, test pixels take the class of their dominant endmember. EM is the angle between learned endmembers and the mean spectra of pure training pixels, paired one-to-one. The true-spectrum reference unmixes noisy test pixels with all measured specimens; it is undefined indoors and for NVIDIA HQ, whose transport the fixed dictionary does not represent.

Further metrics. Reconstruction error stays above the observation noise level on both training and test views, so no model fits the noise (Table S2). The true-spectrum reference reaches 0.84 to 0.94 on clean radiance but drops under noise, most in City and Forbidden City, so both sensor noise and the learned decomposition limit identification.

B. Spectra of SWR-Bench

Sources and selection. Reflectance comes from ECOSTRESS [7] and the USGS Spectral Library 7 [5] (Table S3). A record is accepted only if it is a reflectance measurement that covers 400–1000 nm with gaps of at most 20 nm and values in $[0, 1]$, and if its metadata satisfy a written rule for its class, for example photosynthetic grass but no dry grass, or bark only on trunks. Accepted records are interpolated on a 1 nm grid without smoothing or extrapolation and integrated against 64 Gaussian bands of 20 nm width. Several specimens per class and separate

Table S1. **The eight worlds of SWR-Bench.** Views: training / validation / test. Noise: electrons at the reference radiance and read noise. Obs.: PSNR of the noisy test views against the clean ones.

World	Content	Cameras	Views	Transport	Noise (e^-)	Obs. (dB)
Forbidden City	palace complex inspired by the Forbidden City, with golden roofs, red walls, courtyards, vegetation and water	3 orbits of 60 at 45, 170 and 430 m, pitched down 13°, 32°, 60°	135/21/24	two-term	5,000 / 2	45.4
Market	market square with stalls and pottery, enclosed by terrain, walls and buildings	3 rings of 60, inward and downward	135/21/24	two-term	5,000 / 2	45.6
Botanical	botanical garden with planters, trees and pergolas, enclosed by terrain, walls and buildings	3 rings of 60, inward and downward	135/21/24	two-term	5,000 / 2	46.4
NVIDIA HQ	twin triangular-roof offices, glazed facades, courtyard canopy and connected walkways, parked vehicles and 300 varied trees	descending spiral, facade, courtyard and low campus routes, 60 views each	180/28/32	extended	5,000 / 2	46.7
Study	scholar's study with timber shelves, desk, paper, ink, inkstone and ceramic vessel, 5,000 K ceiling panel	eye-level and elevated orbits, desk details	64/16/16	path traced	3,000 / 3	44.8
Karst	river valley with limestone cliffs, vegetation, soil, wood and clear water	two river-bank routes, a cliff loop and a high loop	128/32/32	two-term + water	5,000 / 2	44.3
City	campus block in a surrounding city: buildings, roads with markings, 297 trees, vehicles	4 elliptical orbits of 60 at 16–95 m	180/28/32	two-term	5,000 / 2	45.2
Office	open-plan office with desks, displays, textiles and plants; 35 LED downlights and daylight through blinds	eye-level and elevated loops, workstation details	64/16/16	path traced	3,000 / 3	46.0

Table S2. **Further metrics of reference Spectral World Renderer on SWR-Bench** (test views, K equal to the number of classes). Train: PSNR on training views. p95: 95th-percentile angle. Acc.: pixel accuracy. Clean: mIoU when trained on clean radiance. GT dict.: NNLS with the true class spectra on noisy / clean radiance.

	PSNR↑					GT dict. mIoU	
World	Train	Test	p95↓	Acc.↑	Clean	noisy	clean
Forbidden City	29.82	28.35	14.76	0.75	–	0.55	0.94
Market	34.11	33.72	8.13	0.66	0.42	0.70	0.94
Botanical	33.11	32.92	10.48	0.65	0.38	0.73	0.93
NVIDIA HQ	30.88	25.20	25.15	0.62	–	–	–
Study	33.09	33.06	7.00	0.89	0.52	–	–
Karst	29.29	28.07	9.73	0.95	0.53	0.77	0.84
City	29.18	28.13	13.58	0.74	0.24	0.32	0.93
Office	29.80	27.23	14.22	0.62	0.29	–	–

spectra under sunlight and skylight give each material a range of appearances (Figure 3 of the main paper).

Realism. We compare every rendered spectrum with independent library measurements of the same type, excluding the rendered records themselves (Table S4). Of 40 natural-material types with enough references, 37 lie within 3° of one and the rest within 4.5°; all 21 vegetation spectra pass red-edge and NDVI checks [3, 11], whereas the dry and foliage-coloured substitutes of the City variants fail them, as designed. Man-made materials are more varied, and pigment proxies for paints lie farthest from their references.

Proxies and limits. Some objects use a measured specimen of a related material, such as pigment powders for paints and road markings, cotton for paper, or granite for an inkstone. Glazing is opaque, most outdoor transport has no spectral interreflection, and the bands end at 980 nm, so short-wave infrared features are out of range.

C. Implementation Details

RGB anchor. The anchor uses multi-view density control [10], which clones and splits Gaussians that cover many high-error pixels and prunes by opacity and multi-view importance. After fitting, colours are discarded and the rasterizer's screen-space variance is folded into the scales,

$$\boldsymbol{\sigma}_i^2 \leftarrow \boldsymbol{\sigma}_i^2 + 0.3\,\mathrm{med}_v(z_{iv}/f_v)^2, \tag{S1}$$

where z_{iv} is the depth of Gaussian i in camera v and f_v its focal length. The harmonic features are then refined for 5k steps without further densification.

Appearance clusters. Each training view contributes up to 2304 smoothed pixel spectra. The descriptor concatenates the unit spectrum with the log norm, robustly standardized and clipped to $[-4, 4]$ and weighted by $\kappa \in [.05, .2]$. Farthest-point seeding and 20 Lloyd steps give K clusters.

Training. All methods train on the noisy training views at 480×270 and are rendered at 960×540 from the same stereo point cloud. The Spectral World Renderer decoder is trained for 30k Adam steps on 2048 pixels of one view each, with learning rate 10^{-3} decayed by a cosine schedule to 10^{-5}, and an EMA with decay 0.99 is evaluated. Endmembers are frozen for 3k updates, then learned at rate 10^{-3}; the assignment weight is $\max(.005, .05(1 - t/3000))$. Reference models without regularization use K equal to the number of classes on SWR-Bench and $K = 16$ on real captures. For 3DGS + VCA/NNLS, a 64-channel 3DGS is trained for 30k steps, VCA [8] extracts K endmembers from its training renders, and each test pixel is unmixed against them by non-negative least squares [6]. UnMix-NeRF uses the official code with its VCA initialization and learning-rate schedule for 30k iterations, accumulating gradients over smaller ray batches; every method trains per scene on the same views.

Dictionary capacity. For every world, K is selected from the class count, 16, 24 and 32 by the dominant-endmember mIoU on the validation views (Table S1), with the regularization and seed 42; in all cases the endmember-to-class map is fitted on the training views. Table S6 lists the test

Table S3. **Material classes and measured spectra of the eight SWR-Bench worlds.** Benchmark class names describe assigned material roles, which may use measured proxies. Counts deduplicate native files; $n_{\mathrm{E}}{+}n_{\mathrm{U}}$ denotes ECOSTRESS and USGS records, a single number ECOSTRESS, and a subscript U USGS. Parent-label counts appear in brackets. Forbidden City uses pigment proxies for yellow tiles and some paints, measured reflectance on opaque water surfaces, and bright solid-material proxies for haze. Karst water instead uses pure-water absorption and a Fresnel interface, with no fixed reflectance endmember. City variants share City geometry, cameras and sensor.

World	Classes	Spectra	E / U	Classes with their number of spectra	Illumination
Forbidden City	9 (11)	47	27 / 20	yellow glazed tile 5_{U}; vermilion paint 3_{U}; vegetation 8; white marble 4; grey masonry 4+5; painted eaves 2+2; soil and bark 7; water 1+2; distant city haze 1+3	G173 sun + sky
Market	6 (8)	29	29 / 0	travertine 1; stone group (white marble, pink marble, granite) 7; vegetation 8; fired clay 4; sandy soil 4; wood and bark 5	G173 sun + sky
Botanical	6 (8)	29	29 / 0	travertine 1; stone group (white marble, pink marble, granite) 7; vegetation 8; fired clay 4; sandy soil 4; wood and bark 5	G173 sun + sky
NVIDIA	8	27	21 / 6	white coating/marking 2+1; glazing 1; vegetation 8; dark panels/frames 2_{U}; concrete 1+2; asphalt 1+1; soil and bark 7; wood 1	G173 sun + sky
Study	5	5	3 / 2	wood 1; paper 1_{U}; ink 1_{U}; dark stone 1; ceramic 1	5000 K panel
Karst	5	10	10 / 0	limestone 2; vegetation 5; soil 1; wood and bark 2; water (absorption model)	G173 sun + sky
City	11	56	33 / 23	concrete/masonry 4+3; soil and bark 6; vegetation 6+1; fired clay 6+2; asphalt + white marking 1+3; yellow marking 2_{U}; dark glazing 2; painted facade 1+5; metal roofing 2+2; vehicle 4+4; dark backdrop 1+1	G173 sun + sky
Office	8	35	11 / 24	light wood 2+4; dark laminate 2+3; textile 5_{U}; grey floor 3_{U}; white paint/plastic 6_{U}; vegetation 4; metal 2+2; display 1+1	LED-B3 + G173
City-metamer	12	59	35 / 24	concrete/masonry 4+3; soil and bark 6; vegetation 6+1; fired clay 6+2; asphalt + white marking 1+3; yellow marking 2_{U}; dark glazing 2; painted facade 1+5; metal roofing 2+2; vehicle 4+4; dark backdrop 1+1; metameric substitute 2+1	as City
City-stress	11	60	36 / 24	concrete/masonry 4+3; soil and bark 6; vegetation 9+2; fired clay 6+2; asphalt + white marking 1+3; yellow marking 2_{U}; dark glazing 2; painted facade 1+5; metal roofing 2+2; vehicle 4+4; dark backdrop 1+1	as City

Table S4. **Reflectance checks per world,** over 420–980 nm. Spectra: distinct measured spectra rendered. Ref.: spectra with at least three independent library measurements of the recorded source type; $< 3^{\circ}$: how many are within 3° of one; med.: median nearest angle. Veg.: green-leaf feature checks. Missing independent references are unevaluated, including Forbidden City pigment-to-tile, opaque-water and distant-background proxies. Counts concern source measurements, not surveyed historical materials. A shared spectrum counts in each world; summaries count each (file,type) pair once (154 pairs of 125 distinct spectra). Natural: vegetation, bark, soil and rock.

World	Spectra	in $[0, 1]$	Ref.	$< 3^{\circ}$	med. ($^{\circ}$)	Veg.
Forbidden City	47	47/47	24	22	1.28	8/8
Market	29	29/29	21	17	2.01	8/8
Botanical	29	29/29	21	17	2.01	8/8
NVIDIA	27	27/27	19	17	1.35	8/8
Study	5	5/5	2	1	6.09	–
Karst	10	10/10	9	9	0.60	5/5
City	56	56/56	30	20	2.07	7/7
Office	35	35/35	18	8	3.46	4/4
Six outdoor worlds	119	119/119	61	47	1.47	17/17
All eight worlds	154	154/154	78	55	2.00	21/21
natural materials	45	45/45	40	37	1.25	21/21
man-made materials	109	109/109	38	18	3.46	–

mIoU of every candidate. The validation choice is also the best test candidate in seven worlds; in NVIDIA HQ, $K = 24$ and $K = 32$ differ by 0.001 on validation and 0.004 on test. The table also gives the decoder without regularization and 3DGS with VCA/NNLS, both at $K = 24$. In Karst and Study, the regularization trades rare classes (soil, ink) for appearance separation of the frequent classes. With $K = 24$, 3DGS with VCA/NNLS improves over its class-count setting in most worlds and remains below Spectral World Renderer in every world except Study, where both reach 0.51–0.52.

D. Decoder Analysis

Fidelity and identification. Table S7 extends the decoder study of the main paper (Sec. 4.2) to a second scene. Replacing unmixing by a free spectral decoder gains at most 0.07 dB, whereas a single feature per Gaussian costs 0.4 to 1.6 dB: fidelity comes from the feature field rather than from relaxing the mixing model. Identification depends on other components. Without harmonic encoding or a separate intensity, Market keeps its PSNR but loses much of its mIoU, as the mixtures absorb shading, and decoding after compositing fits the spectra best but identifies fewer materials.

What the features encode. Projected on their principal components, the features follow both material and shading (Figure S1). Within one Gaussian, intensity varies with shading, while mixture coefficients change mainly at material boundaries, so a single primitive can span an edge. Some mixture variation also follows shadows, one reason a material can occupy several endmembers.

What the decoder learns. Figure S2 compares the decoded quantities with the stored ground truth on test views. Within one specimen, the logarithm of the composited intensity correlates with that of the true radiance ($r = 0.97$ on Market, 0.87 on City), and the sun or shadow state explains 84 % and 67 % of its within-class variance but only 3 % and 15 % of the variance of the class mass. Individual endmembers, in contrast, separate the two states: the endmembers that dominate one Market class in sun and in shadow overlap by 5 %, and those that dominate shadowed surfaces lie closer to

Table S5. **Spectral reconstruction on real captures, per scene** (test views; the main paper reports the means). MS/HS: measured multispectral/hyperspectral; 3DGS stores one channel per band. PSNR is higher-is-better; SAM and p95 angle are lower-is-better. Darker shading marks best, second and third; OOM is retained.

Source	Scene	Spectral World Renderer			HyperGS			DD-HGS			3DGS			UnMix-NeRF		
		PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95
MS	NeSpoF scene1	30.74	3.82	7.12	31.38	4.66	9.46	21.77	3.61	6.65	30.68	3.63	6.82	21.77	13.70	22.79
	NeSpoF scene2	28.38	3.57	8.00	26.56	4.30	9.15	21.40	3.47	7.64	27.57	3.32	7.44	21.69	14.33	22.41
	NeSpoF scene3	29.80	4.19	9.36	26.05	5.11	10.91	25.42	4.91	10.64	25.13	5.40	11.33	20.54	8.90	15.41
	NeSpoF scene4	28.64	3.85	7.66	29.44	4.27	8.24	23.09	3.44	6.82	29.07	3.68	7.41	19.36	11.56	20.40
	Mean	29.39	3.86	8.04	28.36	4.58	9.44	22.92	3.86	7.94	28.11	4.01	8.25	20.84	12.12	20.25
HS	HSI-SC apple	30.67	1.35	3.87	27.95	1.64	4.88	26.69	1.67	5.07	26.29	1.65	5.36	OOM		
HS	Basil	30.30	6.19	17.19	29.58	7.01	18.51	28.32	6.67	17.66	29.99	6.21	17.13	19.42	17.10	34.39
	Rosemary	29.21	4.30	21.48	29.05	4.60	22.35	27.70	6.96	27.18	28.57	4.43	23.31	18.67	15.82	37.50
	Harsh1	21.31	6.22	17.69	20.92	6.81	19.14	20.95	5.99	16.55	21.02	5.99	16.88	16.26	13.28	41.92
	Harsh2	24.78	6.71	17.60	23.07	6.54	16.08	22.34	5.84	14.73	23.09	6.02	15.20	15.34	21.28	51.16
	Fakeplant1	26.93	6.44	12.60	27.86	7.49	15.55	30.12	5.86	11.62	28.93	5.92	11.24	22.33	9.63	21.15
	Mean	26.51	5.97	17.31	26.10	6.49	18.33	25.89	6.26	17.55	26.32	5.71	16.75	18.40	15.42	37.22

Table S6. **Dictionary capacity** (dominant-endmember mIoU, seed 42). Test mIoU of the regularized decoder for each candidate K; shaded: the K selected on validation views, whose validation mIoU is given in column Val. Last two columns: test mIoU at $K = 24$ without regularization and of 3DGS with VCA/NNLS. Karst includes water.

		Regularized, test					$K = 24$, test	
World	Classes	K=classes	16	24	32	Val	w/o reg.	3DGS
Forbidden City	9	0.410	0.500	0.538	0.552	0.563	–	0.277
Market	6	0.340	0.611	0.784	0.754	0.783	0.505	0.323
Botanical	6	0.315	0.628	0.684	0.712	0.713	0.629	0.356
NVIDIA HQ	8	0.202	0.307	0.311	0.315	0.320	–	0.199
Study	5	0.402	0.516	0.521	0.517	0.517	0.588	0.514
Karst	5	0.504	0.562	0.561	0.725	0.727	0.694	0.395
City	11	0.263	0.370	0.477	0.519	0.507	–	0.179
Office	8	0.437	0.500	0.524	0.527	0.533	0.307	0.163

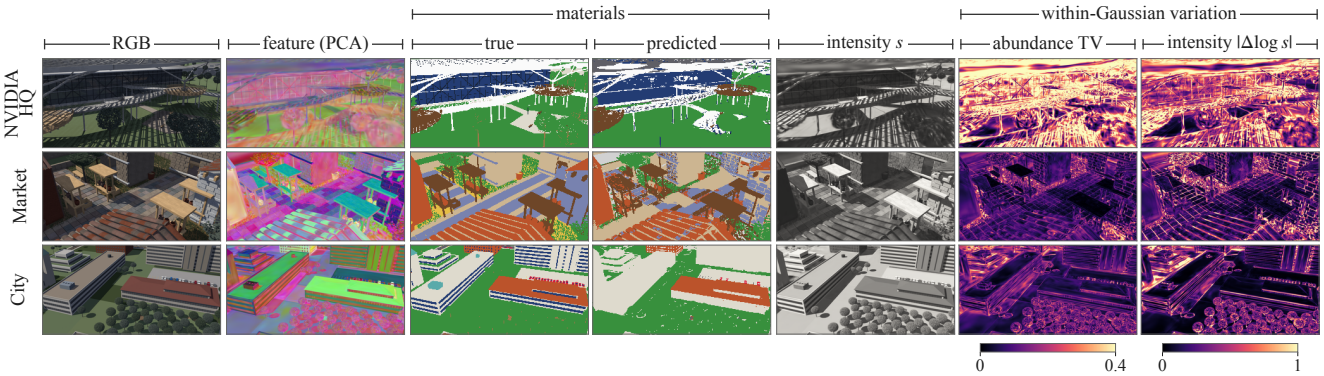

Figure S1. **Harmonic features and their decoding.** Columns show RGB, the per-hit feature (PCA), true and predicted materials, intensity, and the variation of mixture and intensity within each Gaussian, measured against its mean feature at the same viewing direction (fixed scales across scenes).

the class-mean sky radiance than to the direct radiance (5.0° versus 13.2°), so the over-complete dictionary represents illumination modes. In Office, where daylight and downlight differ in spectral composition rather than in brightness, the illumination explains 56 % of the within-class variance of the class mass, and grey flooring near the windows is assigned to white paint. Abundances change little with the viewing direction on the Lambertian worlds (TV at most

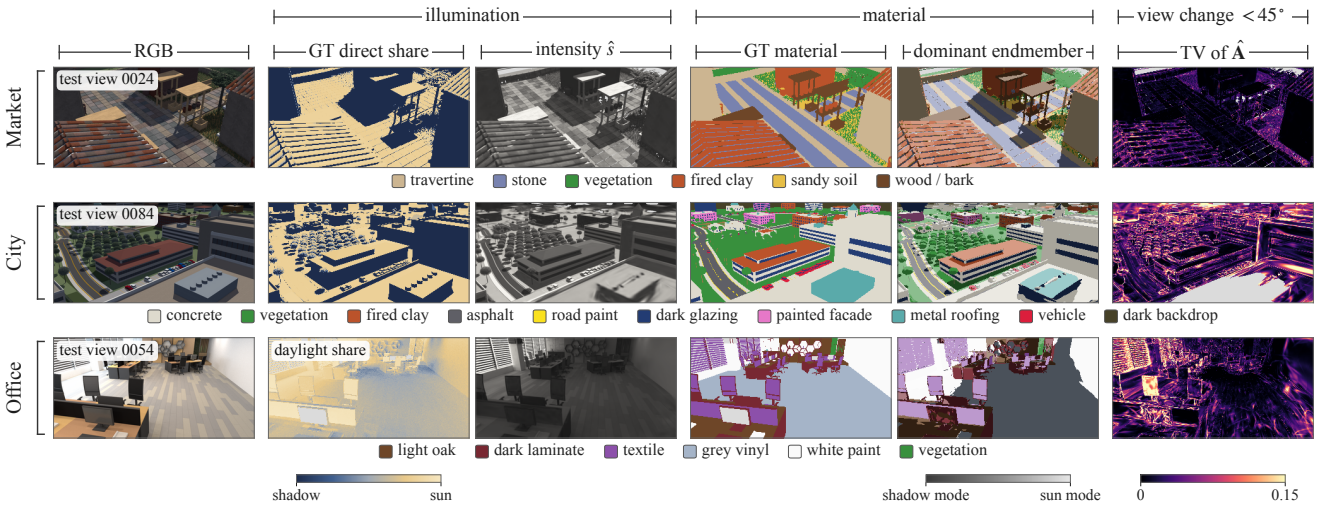

Figure S2. **Intensity carries shading; endmembers carry material and illumination mode.** Test views of Market (K=24), City (K=32) and Office (K=32). The composited intensity $\hat{s}$ follows the direct share of the illumination, and each material occupies distinct endmembers in sun and in shadow, so the over-complete dictionary represents illumination modes within a class. Columns: RGB; ground-truth direct share (Office: daylight share); $\hat{s}$; ground-truth material; dominant endmember, coloured by its class and shaded by the mean direct share of its pixels; total variation (TV) of $\hat{\mathbf{A}}$ when the viewing direction changes by up to 45°.

Table S7. **Decoder controls.** One change at a time, without appearance regularization; means of seeds 42 and 43. Free decoders output spectra.

	NeSpoF scene3		Market		
Variant	PSNR↑	SAM↓	PSNR↑	SAM↓	mIoU↑
Per-hit unmixing	29.62	4.22	33.72	1.87	**0.47**
One feature/Gaussian	29.24	4.30	32.09	2.08	0.34
No harmonic encoding	29.61	4.21	33.65	1.89	0.30
Deferred (pixel)	29.16	**4.17**	34.37	1.57	0.37
No intensity (s=1)	29.54	4.37	33.70	2.14	0.22
Free, per hit	**29.69**	4.28	33.74	1.72	–
Free, one feature/Gaussian	29.26	4.34	32.11	1.90	–
Free, deferred (NHT)	29.21	**4.17**	**34.38**	**1.41**	–

0.04 within 45°), whereas the NVIDIA HQ decoder depends on the viewing direction throughout (0.18), not only on glass and specular surfaces.

Primitives as material carriers. Figure S3 draws the tetrahedron that bounds each Gaussian and colours it by the class decoded at its centre. Tetrahedra of one surface share a class, and class boundaries in the rendered map coincide with boundaries between primitives, so the abundances that a material query selects belong to individual Gaussians.

E. Real Captures

Data. HS-NeRF cubes are aligned with the released RGB images and COLMAP cameras, and only uniquely aligned views are kept; Fakeplant2 and Pinecone are excluded, as their processed RGB has a white background absent from the raw cubes. Figures S4–S6 show the first test view of every scene; errors concentrate at edges and thin parts.

Table S8. **Real captures.** Views are training/test views per scene; SO/BS denote Surface Optics/BaySpec sensors.

Dataset	Scenes	Bands	Range (nm)	Resolution	Views
NeSpoF	4	21	450–650	512 × 428	15–26 / 3–4
HSI-SC apple	1	204	397–1004	512 × 512	56 / 8
HS-NeRF (SO)	4	128	371–1044	348 × 260	20–42 / 3–7
HS-NeRF (BS)	1	141	401–1087	320 × 256	50 / 8

Table S9. **City training time by stage** (one RTX 4090, seed 42, including start-up and loading; ours: mean of two runs).

Method	Stage	City (s)
UnMix-NeRF	VCA initialisation and training	1,457
3DGS + VCA/NNLS	64-band 3DGS	5,030.2
	rendering and VCA	71.3
	total	5,101.5
SWR	anchor Gaussians	72.8
	harmonic features	36.5
	appearance clusters	68.4
	decoder	219.5
	total	**397.2**

F. Efficiency

Storage. Each Gaussian stores 59 values at any band count, against 78 for HyperGS and $62{+}L$ for per-band fields, so our models are 6 to 32 times smaller on the real captures, for example 4.2 MB against 127 MB on the 204-band apple.
Training and rendering. The City pipeline takes 397 s (Table S9); anchor and features do not depend on K, so tuning

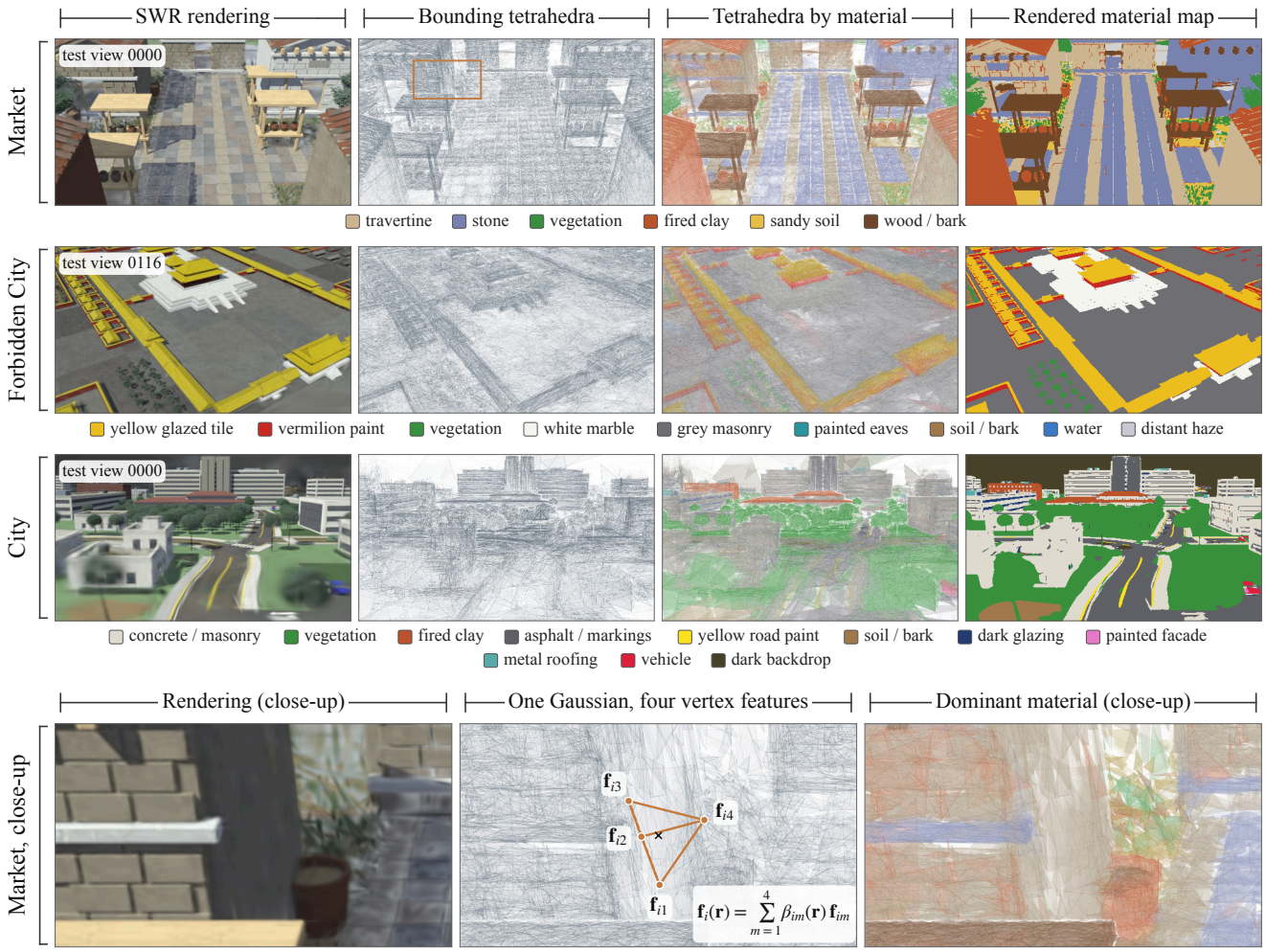

Figure S3. **Bounding tetrahedra on test views.** Each Gaussian G_i is enclosed by the regular tetrahedron of its canonical frame, scaled by $\boldsymbol{\sigma}_i$, rotated by $\mathbf{R}_i$ and translated by $\boldsymbol{\mu}_i$; the barycentric coordinates of the point of maximal response weight its four vertex features (Eq. 3 of the main paper). Columns show the rendering, the tetrahedra weighted by opacity and visibility, the tetrahedra coloured by the dominant class decoded at each Gaussian centre, and the rendered material map. The tetrahedra follow the scene structure, and their colouring reproduces the class layout of the rendered map: the median visible Gaussian places more than 98 % of its centre abundance on one class. The bottom row isolates one Gaussian of Market (amber window).

Table S10. **Rendering speed** (one RTX 4090): test views per second at $960 \times 540 \times 64$, including spectra and labels. For 3DGS + VCA/NNLS, the second row adds the training renders that VCA needs. Ours: full precision (reference), fp16 skipping intersections of weight $< 10^{-4}$ (fast), and the same in one fused kernel; dominant labels agree with the reference on 99.97 % of pixels.

Method	Market	Botanical	Study	Karst	Mean
UnMix-NeRF	0.18	0.21	0.21	0.12	0.18
3DGS + VCA/NNLS	2.53	2.51	4.70	4.73	3.6
incl. training renders	1.49	1.47	2.18	2.16	1.8
SWR, reference	3.38	3.79	3.52	2.64	3.3
SWR, fast	8.19	8.81	8.09	6.34	7.9
SWR, fused	**36.5**	**39.3**	**36.3**	**27.4**	**34.9**

K repeats only the clustering and the decoder. Decoding every intersection costs more than rasterizing a stored spectrum, but our fused fp16 kernel, which skips intersections of negligible weight, renders 34.9 labelled views per second, about ten times 3DGS with VCA (Table S10).

G. Rare Materials and Altered Leaves

Rare classes. Class IoU falls with class frequency, while the true spectra still separate most rare classes, as Figure S7 shows; the errors therefore lie in endmember allocation rather than in the data.

Altered leaves. In the City variants, foliage-coloured mineral leaves and stressed leaves are separated from healthy

Figure S4. **Qualitative comparison on HS-NeRF test views**, the first test view of each of the five scenes. Rows show false colour (630, 550 and 460 nm) and per-pixel spectral angle (left scale). Bold marks the best scores.

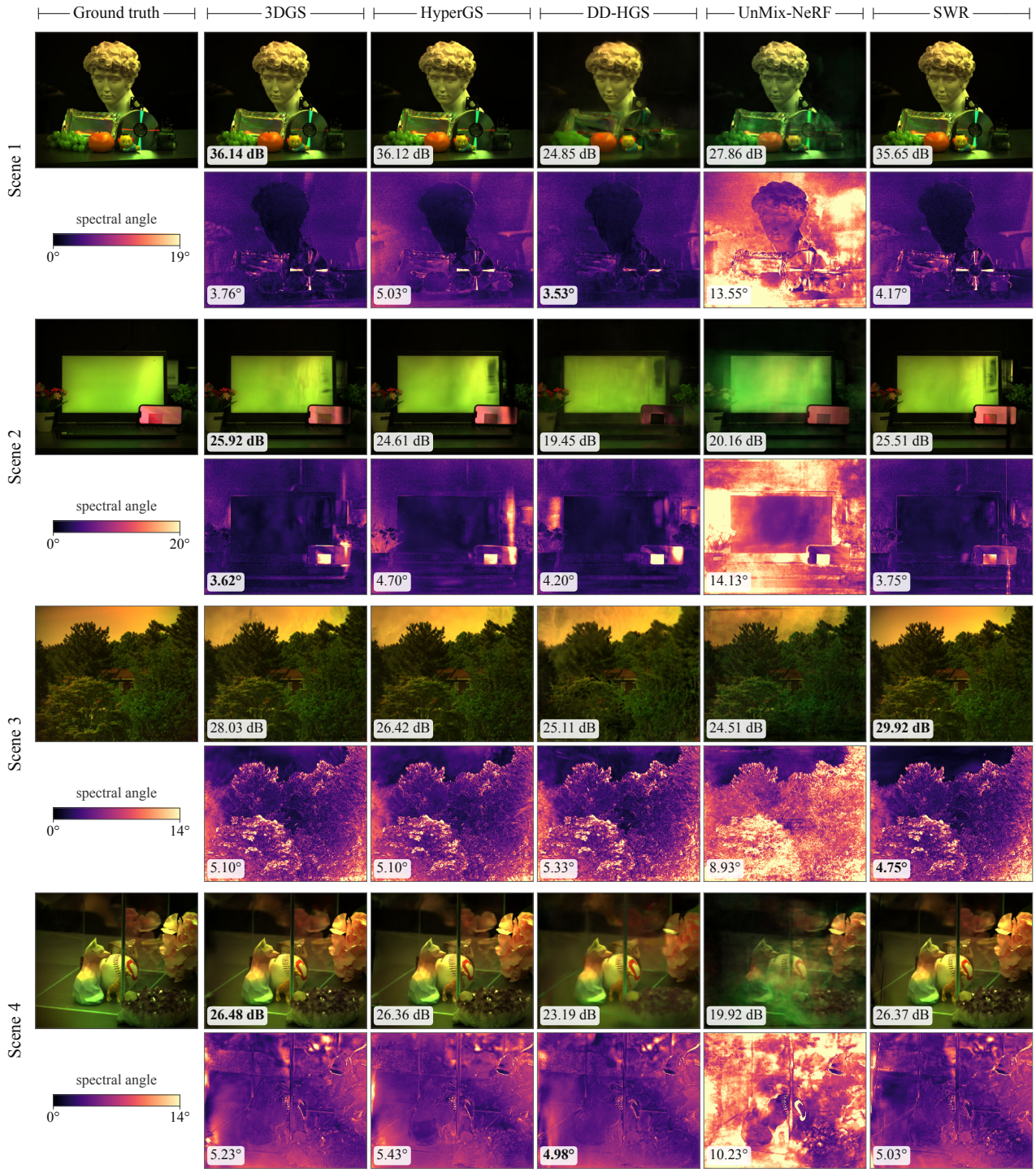

Figure S5. **Qualitative comparison on NeSpoF test views**, the first test view of each of the four scenes, as in Figure S4.

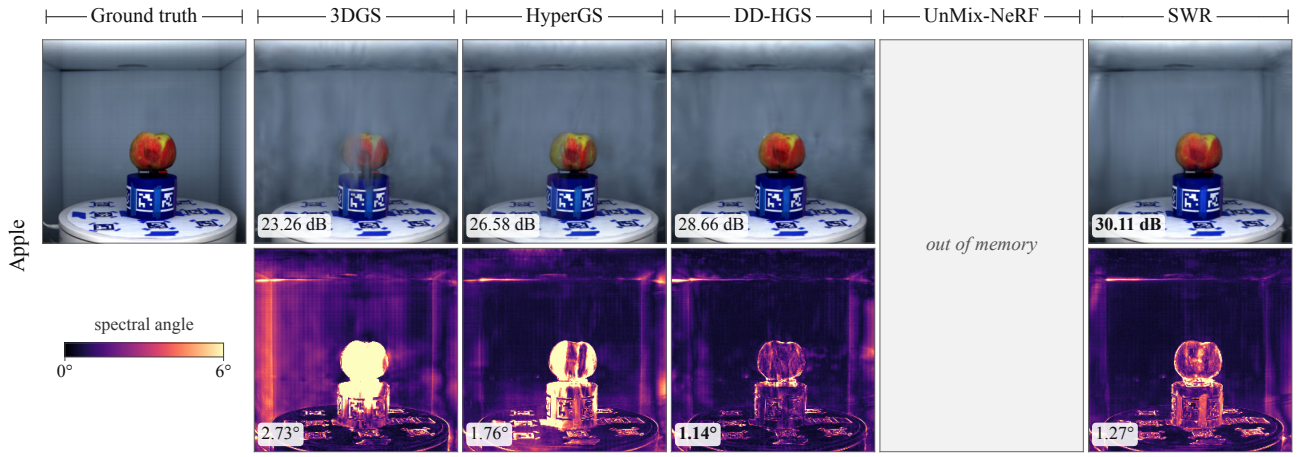

Figure S6. **Qualitative comparison on the HSI-SC apple**, first test view, as in Figure S4. UnMix-NeRF runs out of memory.

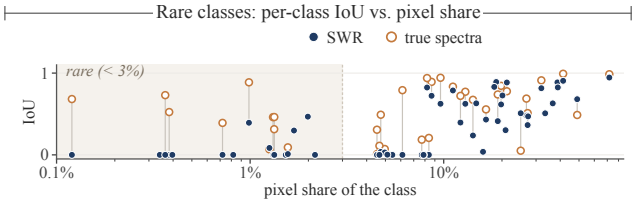

Figure S7. **Rare classes are absorbed by larger ones.** Per-class IoU against pixel share over the eight worlds, with K equal to the class count. NNLS with the true class spectra separates most rare classes that Spectral World Renderer merges into larger ones.

foliage far better by Spectral World Renderer than by clustering RGB: the total variation between their label distributions is .85 and .83, against .10 and .40 for RGB k-means and .95 and .97 for the true spectra (main paper, Figure 9(c)). Separation is not yet identification: most changed leaves are mapped to another class rather than to a class of their own. A moved sun leaves large classes unchanged.

H. Object Queries

Procedure. Each Gaussian is assigned its mean abundance and radiance over the training views. The type "tree" denotes crown materials and a size: Gaussians of crown material above the ground are clustered with DBSCAN [2], and large clusters form instances. A measured leaf spectrum then selects instances by comparing each crown with the query and with ordinary leaves under sunlight and skylight. All parameters were tuned on the base City.

Results. The substituted mineral is 42° from ordinary leaves in spectrum but only 2° in RGB, so RGB retrieves wrong trees and Spectral World Renderer none (Table S11; main paper, Figure 9(a,b)). The true spectra give the same answers, and the missed targets are lost earlier, mostly in crowns too sparse to form instances.

Table S11. **Object queries in City.** Every target is either lost before the spectral query or found; among the found trees, Spectral World Renderer retrieves no ordinary tree in the metamer variant, whereas RGB retrieves up to 11. Targets: substituted (metamer) or stressed trees. Lost: typed as another material, or too few Gaussians to form an instance. Instances: all tree instances returned by the type query. Bold: fewest wrong answers.

		Lost				SWR		RGB	
Seed	Targets	typed	sparse	Found	Inst.	correct	wrong	correct	wrong
Metamer: foliage-coloured mineral leaves									
42	61	13	43	5	57	5	**0**	5	11
43	61	16	43	2	54	2	**0**	2	2
Stress: dry leaves on 87 trees									
42	87	6	62	19	63	19	**1**	19	**1**
43	87	7	62	18	60	18	**1**	18	**1**

I. Query by Spectrum

Procedure. The query is a spectrum: the rendered spectrum at a clicked pixel or a library reflectance $\boldsymbol{\rho}$. Each Gaussian carries its mean spectrum over the training rays, $\bar{\mathbf{h}}_i \propto \sum_{\mathbf{r}} w_i(\mathbf{r})\,\mathbf{h}_i(\mathbf{r})$. Let $\boldsymbol{\phi}_\mathrm{d}, \boldsymbol{\phi}_\mathrm{s} \in \mathbb{R}^L$ denote the direct and sky irradiance of the world integrated over the band responses. Under sun and sky, a surface yields radiance proportional to $\boldsymbol{\rho} \odot (\alpha\boldsymbol{\phi}_\mathrm{d} + \boldsymbol{\phi}_\mathrm{s})$, whose shape varies with the direct shading α. The comparison removes this dependence in one of two ways:

- a library spectrum is compared by its angle to the non-negative cone spanned by $\boldsymbol{\rho}\odot\boldsymbol{\phi}_\mathrm{d}$ and $\boldsymbol{\rho}\odot\boldsymbol{\phi}_\mathrm{s}$, which contains every sunlit, penumbral and shadowed mixture;
- a clicked spectrum is compared after illumination whitening: the RMS log-ratio of the two spectra once the constant vector and the two principal directions of $\log(\alpha\boldsymbol{\phi}_\mathrm{d}+\boldsymbol{\phi}_\mathrm{s})$, $\alpha \in [0,1]$, are projected out. These two directions account for 99.999 % of the variation of this family.

Neither step involves training. The per-Gaussian angles are rasterized with the rendering weights; a default threshold

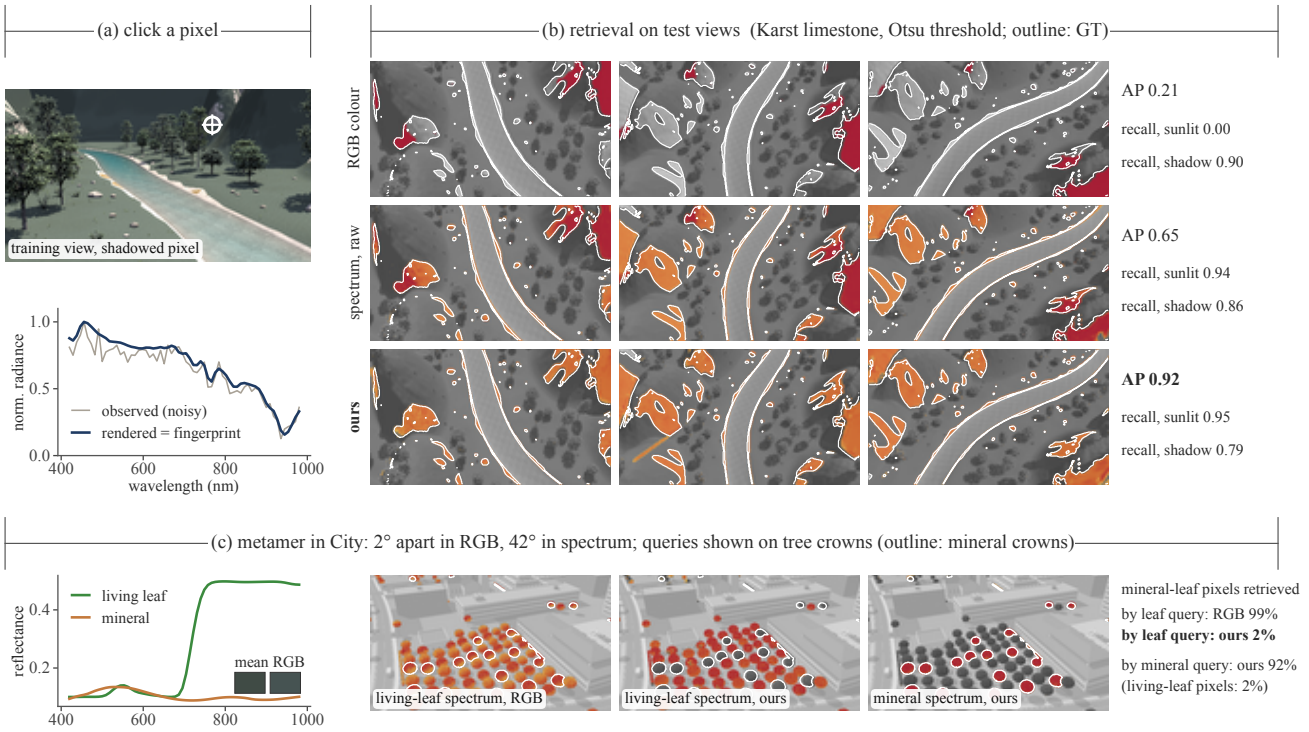

Figure S8. **A clicked spectrum retrieves its material across shading and separates metamers.** (a) A pixel on shadowed limestone is clicked in a training view of Karst; its rendered spectrum is the query. (b) Selections on three test views at the Otsu threshold for the RGB colour angle, the raw spectral angle and the whitened spectral angle on per-Gaussian spectra (white outline: ground truth); AP and recall are computed over all test views for this click. The RGB query misses the sunlit rock, which the whitened query retrieves. (c) City metamer, shown on tree crowns (outline: mineral crowns): the living-leaf spectrum retrieves the mineral crowns in RGB but not with Spectral World Renderer, and the mineral spectrum retrieves only them.

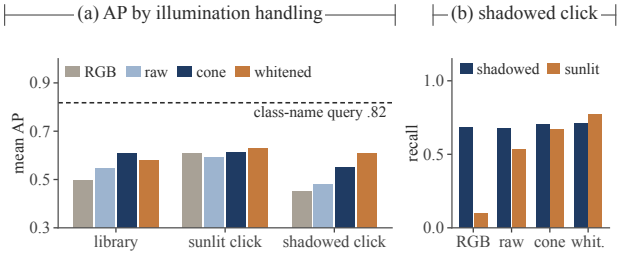

Figure S9. **Illumination handling makes spectral queries robust to shading** (mean over four worlds). (a) Mean AP of RGB queries and of spectral queries without illumination handling (raw), with the sun and sky cone and with whitening; dashed line: class-name query. (b) For shadowed clicks, the RGB query retrieves the shadowed part of the class but almost none of its sunlit part, whereas whitened spectral queries retrieve both.

follows from Otsu's method [9] on the histogram of log angles, and a slider adjusts it.

Evaluation. Market, City, Botanical and Karst use the models of Table 2. Clicks are three sunlit and three shadowed pixels inside a single class in training views. Library queries use the assigned specimen of each class, another specimen of the class and ECOSTRESS spectra [7] that do not occur in the world. Each query is scored against the class mask on all test views, and the four worlds are weighted equally.

Table S12. **Query by spectrum** (mean over Market, City, Botanical and Karst). AP against the class masks on test views. Recall: proportion of the sunlit part of the class retrieved by shadowed clicks at the Otsu threshold. Ours: illumination cone for library spectra, whitening for clicks; click queries operate on rendered pixels, and per-Gaussian queries differ by at most .02 AP.

	AP↑			Recall↑
Query	library	sunlit click	shadowed click	sunlit part
RGB colour angle	.50	.61	.45	.10
Spectral angle, raw	.55	.59	.48	.53
Ours	**.61**	**.63**	**.61**	**.77**
Class name (reference)	.82	–	–	–

Results. Illumination handling makes a measured spectrum an effective query (Table S12, Figure S9). For shadowed clicks, the spectral query raises AP from .45 (RGB) to .61 and retrieves 77 % of the sunlit part of the clicked material, against 10 % for RGB, as shown in Figure S8(b). For library spectra, the illumination cone raises AP from .50 to .61. For sunlit clicks, spectral and RGB queries perform on par (.63

and .61). Applying the same query to the observed test spectra raises AP by only .03. When the material name is known, the class-name query remains more accurate (.82); a single spectrum also cannot represent classes assembled from several differently coloured specimens, such as vehicles and painted facades. The Otsu threshold yields an IoU about .14 below the best threshold of each query and is unreliable for rare materials, so the slider sets the final selection.

Metamers. In the City variant, the substituted mineral is 2° from foliage in RGB and 42° in spectrum. A living-leaf library spectrum separates the two leaf types with an AUC of .99 on per-Gaussian spectra, against .26 for RGB; it retrieves 2 % of the mineral-leaf pixels, against 99 % for RGB, as shown in Figure S8(c). The mineral spectrum retrieves 92 % of mineral-leaf and 2 % of living-leaf pixels.

Speed. With 61k to 84k Gaussians, a query takes 1.6 to 2.9 ms on an RTX 3090; rasterizing the selection at 960×540 brings the total to 3 to 4 ms.

J. Additional Views of SWR-Bench

Figures S10 and S11 show the simulated hyperspectral observations of one test view per world. Figures S12–S14 show each world from fifteen cameras along its trajectory. Figures S15–S18 then show three test views of each world with predictions. Each row shows the rendered input, the material ground truth and the Spectral World Renderer prediction at one camera, using the models of the main table and their frozen training-view maps. Views are chosen by camera separation from a fixed reference camera, not by prediction scores.

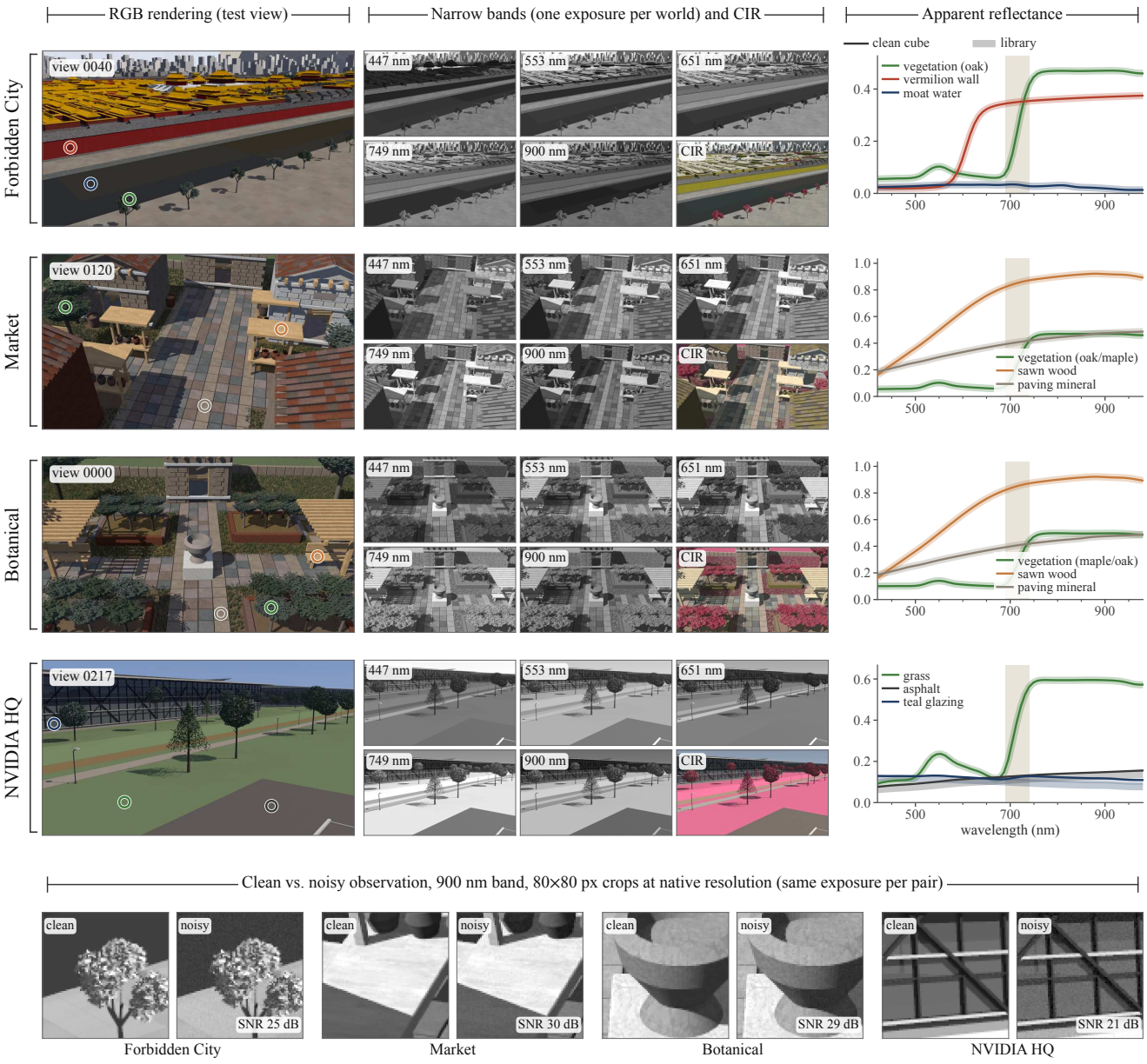

Figure S10. **Simulated hyperspectral observations: Forbidden City, Market, Botanical and NVIDIA HQ.** Narrow bands share one exposure per world, so brightness differences between bands follow the radiance; the colour-infrared composite (R, G, B = 847, 651, 553 nm) renders vegetation red. Apparent reflectance divides the clean radiance by the white-reference radiance under each pixel's illumination (median and 10–90 % range, over the library reflectance of the assigned specimen; shaded: red edge, 690–740 nm). Outdoor spectra reproduce the library reflectance, including chlorophyll absorption near 670 nm and the red edge; path-traced interiors deviate through interreflection and residual Monte Carlo variance. Bottom: clean and noisy crops of the 900 nm band with the signal-to-noise ratio of the noisy crop.

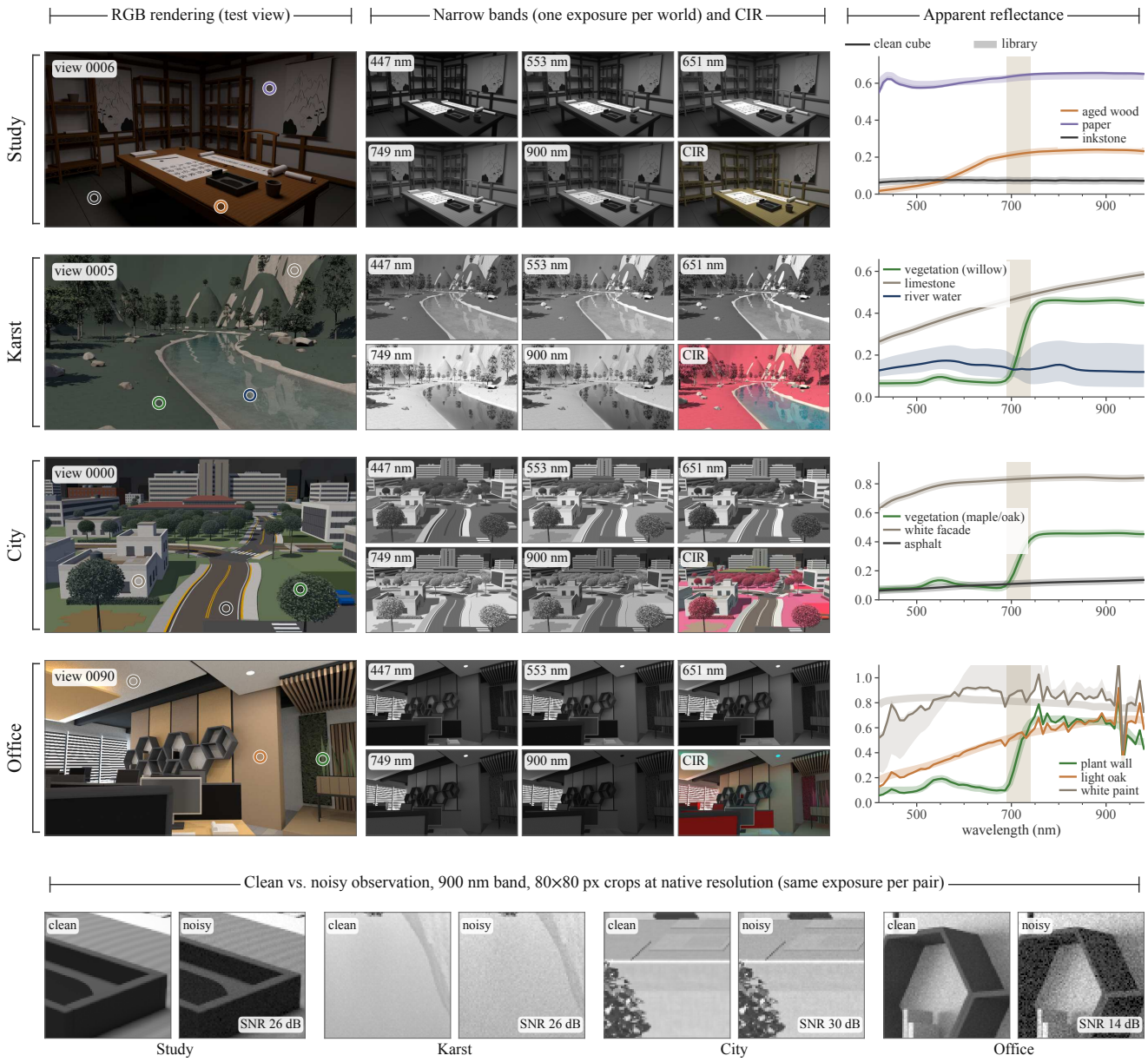

Figure S11. **Simulated hyperspectral observations: Study, Karst, City and Office.** Layout as in Figure S10. Indoor worlds are lit by white sources, so their bands differ less; the Office lights emit no near-infrared, which comes only from daylight through the blinds.

Figure S12. **Rendered views: Forbidden City, Market and Botanical.** Fifteen cameras per world, spread along its trajectory (released RGB renders).

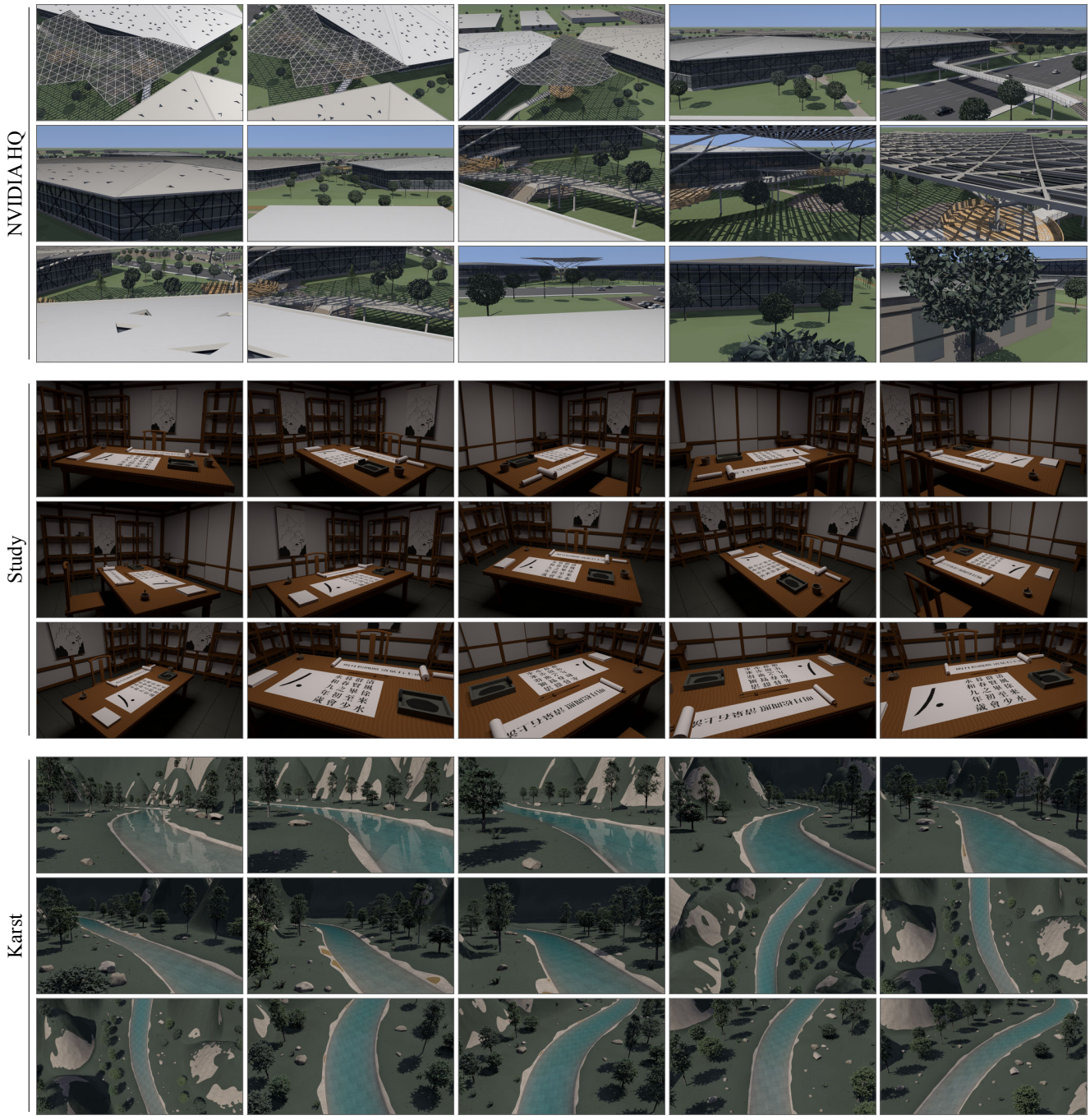

Figure S13. **Rendered views: NVIDIA HQ, Study and Karst.** Fifteen cameras per world, spread along its trajectory (released RGB renders).

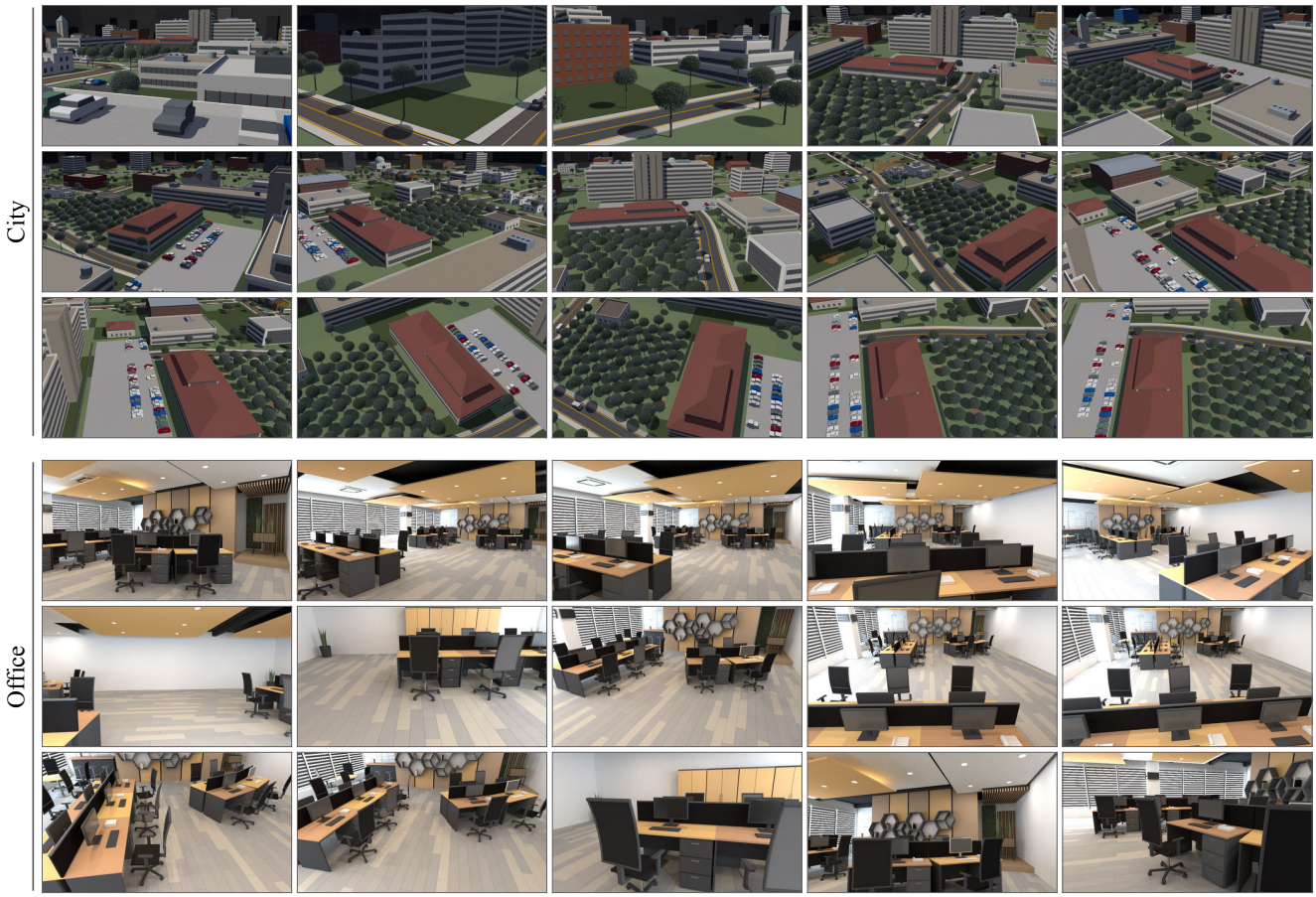

Figure S14. **Rendered views: City and Office.** Fifteen cameras per world, spread along its trajectory (released RGB renders).

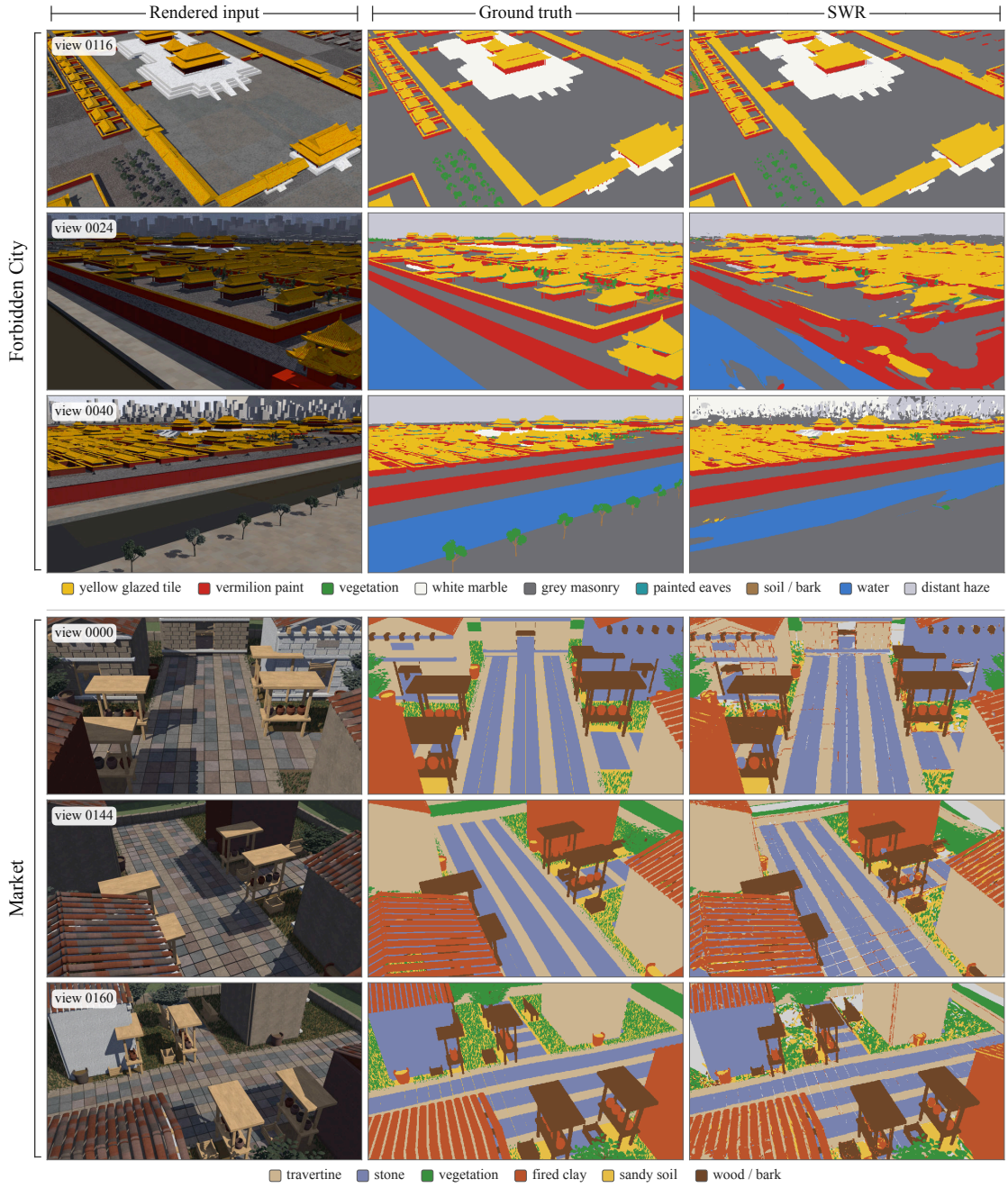

Figure S15. **Additional test views: Forbidden City and Market.** Three separated test cameras per world, shown as full frames.

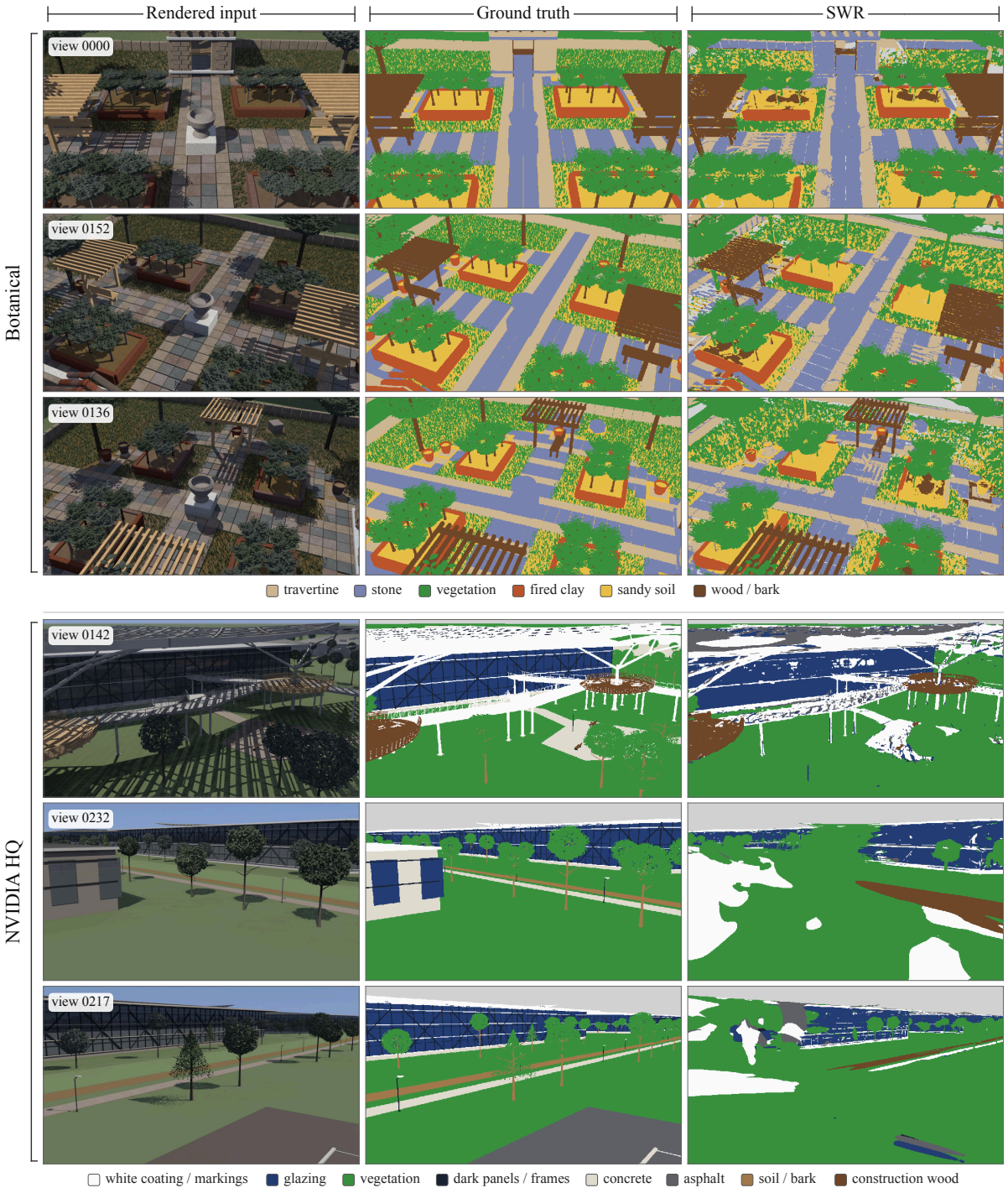

Figure S16. **Additional test views: Botanical and NVIDIA HQ.** Three separated test cameras per world, shown as full frames.

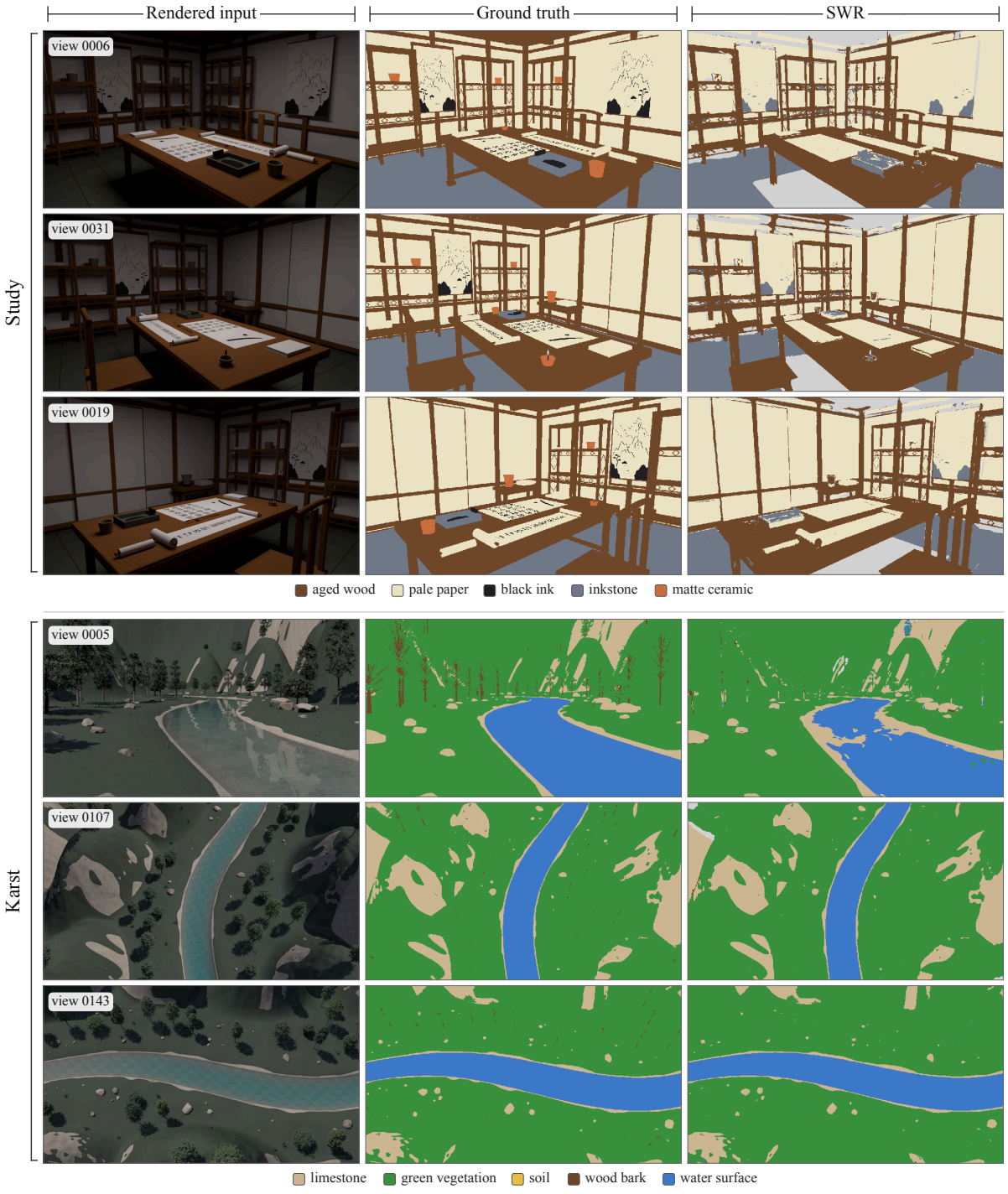

Figure S17. **Additional test views: Study and Karst.** Three separated test cameras per world, shown as full frames.

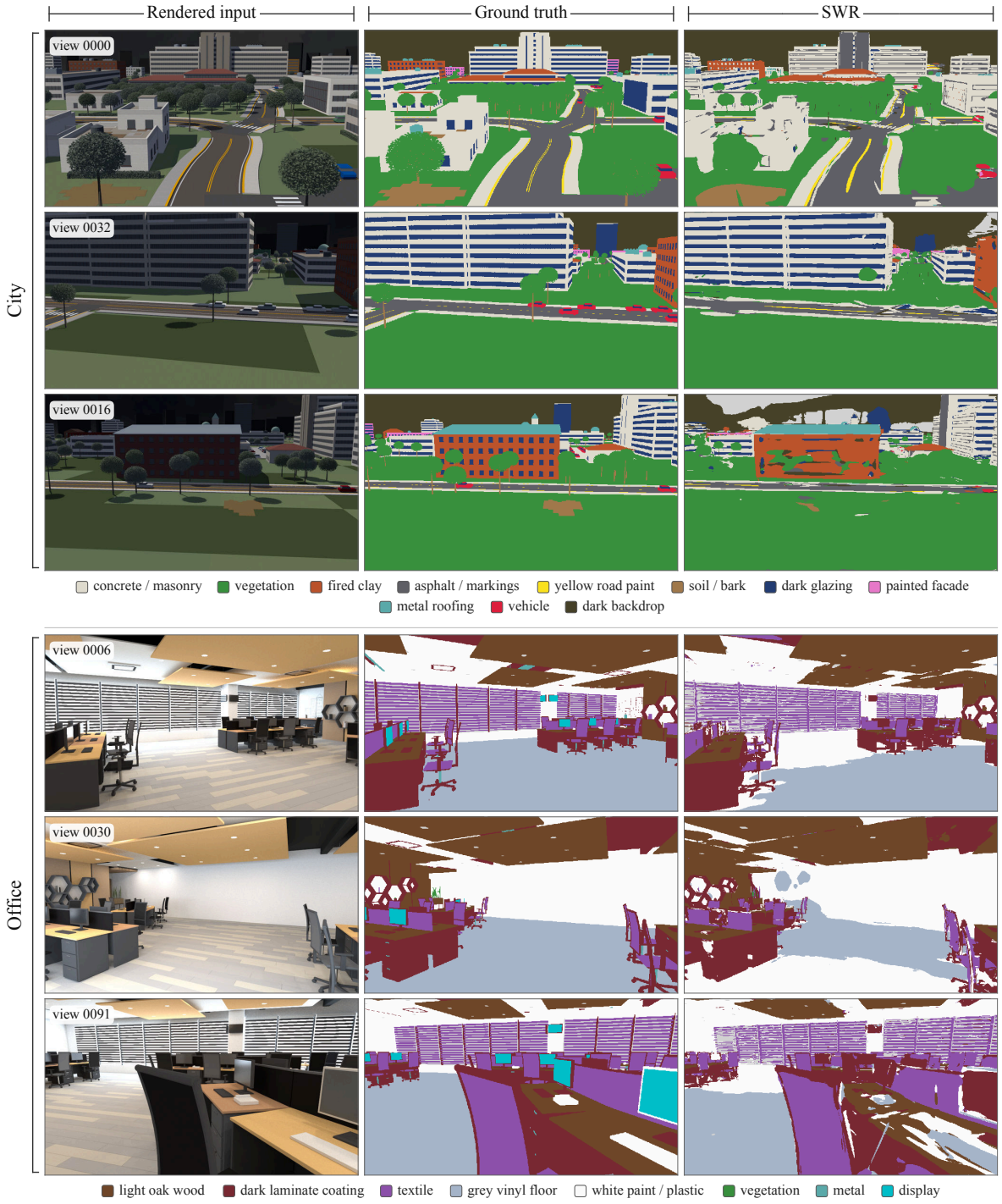

Figure S18. **Additional test views: City and Office.** Three separated test cameras per world, shown as full frames.

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